From Existing Methods to New Research Ideas: Two Perspectives on Recursive Self-Improvement

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Recursive self-improvement (RSI) offers a route to AI systems that participate in their own continued development. Recent methods allow language agents to retain changes to their parameters, memory, skills, and code for subsequent execution and improvement. We examine how these methods can inform new research ideas through two complementary perspectives. First, we introduce a unified taxonomy with nine design dimensions organized into four groups: update targets, update organization, update schedule and scope, feedback and selection. This taxonomy places existing methods in a shared design space, where examining individual dimensions and their interactions can inspire new research ideas. Second, we examine how existing RSI methods draw on traditional deep learning, for example through optimization, regularization, data augmentation, continual learning, and adversarial learning. We develop further ideas by examining more specific learning mechanisms and drawing on evolutionary computation. We also review how existing benchmarks assess self-improvement under online, offline, and offline-to-online settings. Together, these perspectives help researchers develop new RSI research ideas from existing designs and learning mechanisms. RSI papers and resources are available at `https://prism-shadow.github.io/awesome-rsi/`.

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1 Introduction

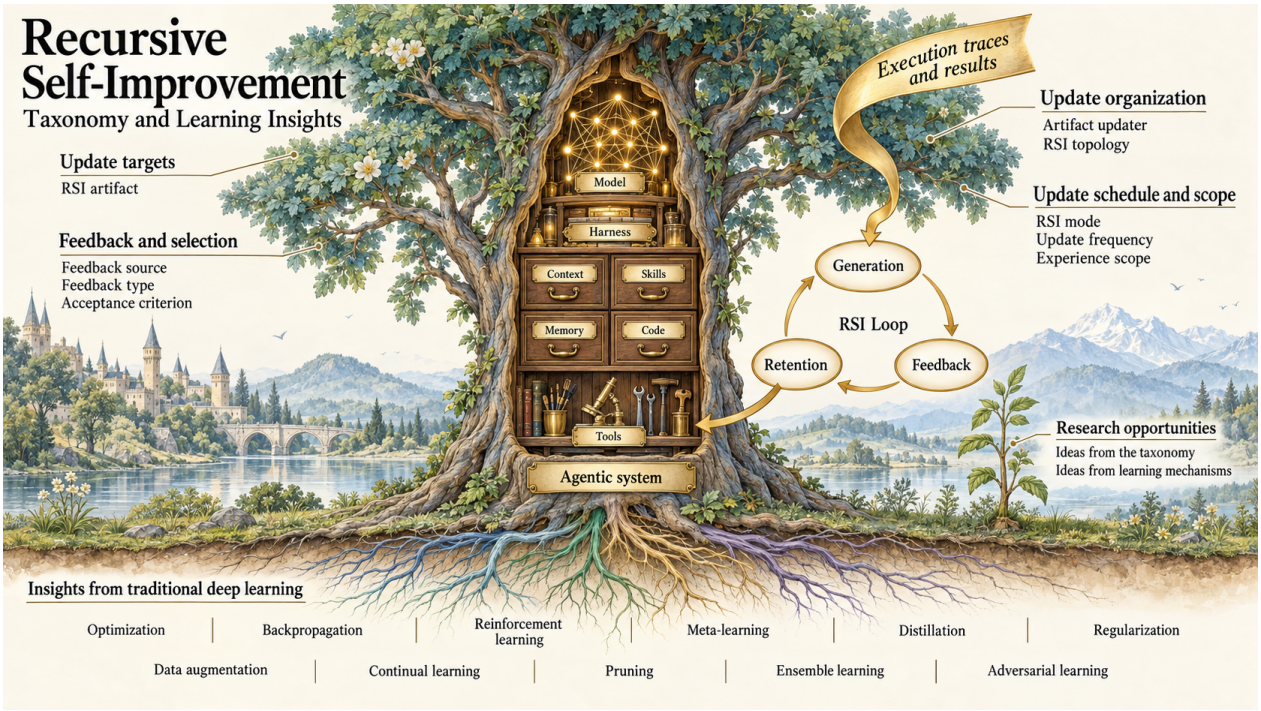


Figure 1 Overview of the survey. Agent resources form the trunk, four groups of design dimensions organize the canopy, and traditional deep learning supplies the roots. Execution traces and results inform self-improvement. New growth represents research opportunities arising from the taxonomy and learning mechanisms.

Recursive self-improvement (RSI) is a process in which an AI system participates in modifying itself, retains these modifications, and uses the revised system in subsequent execution and improvement. Its appeal lies in the possibility that an improvement makes the system better at producing further improvements. A growing share of AI development could then be carried out by the systems being developed. This possibility underlies Good's suggestion that a sufficiently capable machine might be the last invention humans need to make [1].

Recent research has produced numerous RSI methods, including those that update model parameters, revise memory and skills, and modify agent code [2, 3, 4, 5]. Several surveys now organize this literature by improvement targets, autonomy levels, or the degree of loop closure [6, 7, 8]. These classifications help describe the field, but assigning a method to a broad category does not fully explain its design. Two methods that revise agent code, for example, can differ in who proposes changes, how earlier versions are reused, and what feedback determines which changes are retained. Generating new research ideas from existing work requires examining its design dimensions and their interactions to identify limitations of current mechanisms and new ways to improve them. Existing surveys also discuss connections to earlier learning paradigms, but offer less systematic guidance on using specific mechanisms from traditional deep learning to generate new RSI research ideas. This calls for an analysis of how RSI adapts these mechanisms and which principles can be reused.

We view existing RSI methods as combinations of design choices. These choices define a high-dimensional design space: each dimension captures an aspect of the improvement process, and each method occupies a point determined by its choices along these dimensions. Locating existing work in this space allows us to examine individual dimensions and their interactions, identify limitations of current mechanisms, and develop new research ideas for agent improvement. To characterize this space, we introduce a unified taxonomy with four groups of design dimensions: *update targets*, *update organization*, *update schedule and scope*, and *feedback and selection*. These groups encompass nine dimensions, which we use to describe representative methods, allowing methods developed for different settings to be compared within the same space. The accompanying

Awesome RSI website lets readers browse and filter methods along these dimensions.[1]

Traditional deep learning provides a second source of research ideas for RSI. It has developed mechanisms for choosing updates, learning from feedback, and controlling overfitting, all of which are relevant to recursive self-improvement. Recent work already draws on these connections: SkillAdam adapts ideas from Adam to skill revision; TextGrad applies the idea of backpropagation to textual feedback; and RRSI regularizes changes to agent code [4, 9, 10]. We study these correspondences to understand which principles carry over to RSI. We then examine more specific mechanisms within these connections and draw on evolutionary computation to develop further research ideas.

We also review benchmarks and evaluation settings for online, offline, and offline-to-online improvement. This accompanying analysis explains what each setting measures and motivates shared evaluation protocols.

Our contributions are twofold.

- **A unified taxonomy of agent self-improvement.** We introduce nine design dimensions organized into four groups, grounded in a formal description of the self-improvement loop. This taxonomy places existing methods in a common design space, allowing us to compare their mechanisms and examine relationships across dimensions. Through a concrete example, we illustrate how this analysis can reveal limitations of existing designs and inspire new RSI research ideas.
- **Insights from traditional deep learning.** We analyze how existing RSI methods draw on established learning mechanisms, for example through optimization, regularization, data augmentation, continual learning, and adversarial learning. We explain the problems these mechanisms originally address and how RSI methods adapt them to self-improvement. Building on this analysis, we explore how deeper developments of these mechanisms, together with ideas from broader machine learning and evolutionary algorithms, can generate further RSI research ideas.

2 Background

We consider an agentic system to perform recursive self-improvement (RSI) when it modifies itself, retains the changes, and uses the updated system to execute subsequent tasks and produce further self-modifications. We first introduce the agentic system and its task-execution loop, then formalize the self-improvement loop and connect it to the design dimensions developed in our taxonomy.

2.1 Agentic systems

An agentic system consists of a language model and a harness. The harness comprises tools, context, memory, skills, and executable code. Tools are callable operations through which the system obtains information and acts on the environment. Context is the information assembled for a model call, including instructions, task inputs, and relevant interaction history. Memory stores information beyond the current context for retrieval and reuse. Skills package task-specific instructions with optional scripts and reference material as reusable resources. Executable code coordinates model calls, context construction, tool execution, and the use of memory and skills. The environment refers to the external systems that the agent can observe and affect.

2.2 The ReAct Loop

A basic chatbot produces an answer from a query, the supplied context, and knowledge stored in its parameters. An agent extends this setting by taking actions and using the resulting observations to decide what to do next. It can obtain information, execute operations, and revise its plan as the task proceeds. Results can also guide revision of an attempted solution [11]. ReAct organizes this interaction through the interleaving of reasoning and acting [12]. We use *ReAct Loop* to describe this task-execution cycle.

In each iteration, the harness assembles the context for a model call and executes the tool operations requested by the model. Observations from the environment become part of the information available for subsequent decisions. Repeated interaction produces execution traces and task outcomes that can inform changes to the system itself.

[1] GitHub repository: `https://github.com/Prism-Shadow/awesome-rsi`.

2.3 The RSI Loop

The *RSI Loop* connects self-modification to subsequent execution and improvement. The system gathers information, constructs candidate changes, and incorporates selected changes into the configuration used in later rounds. Task execution supplies information for this process.

We formalize the loop to describe how the current system gives rise to the system used in the next round. Let $\mathcal{A}_t$ denote the current system configuration. The update procedure U covers a complete round of self-improvement:

$$U\big(\mathcal{A}_t; D\big) \longrightarrow \mathcal{A}_{t+1}, \qquad t = 0, 1, 2, \ldots. \tag{1}$$

Here, t counts improvement rounds, and D collects the method's nine design settings. Within U, the functions P, F, and R perform candidate generation, candidate feedback, and retention in sequence. Their connection can be written as

$$U(\mathcal{A}_t; D) = R(\mathcal{A}_t, (F \circ P)(\mathcal{A}_t))\,. \tag{2}$$

The settings of P, F, and R are omitted in this overview and specified below. The composition $F \circ P$ generates a candidate and returns it together with its feedback. R uses this pair and the current system to determine the system retained for the next round.

System configuration. Following the components described in Section 2.1, we write

$$\mathcal{A}_t = (\mathcal{M}_t, \mathcal{H}_t), \tag{3}$$

where $\mathcal{M}_t$ contains the system's language models and their parameters. The harness resources are

$$\mathcal{H}_t = \big(\mathrm{Tools}_t, \mathrm{Context}_t, \mathrm{Memory}_t, \mathrm{Skills}_t, \mathrm{Code}_t\big). \tag{4}$$

The entries correspond to the tools, context, memory, skills, and executable code used in round t.

Update schedule and scope. The first three entries of D specify when U operates and where its retained changes are intended to be reused:

$$D = \big(d_{\mathrm{mode}}, d_{\mathrm{frequency}}, d_{\mathrm{scope}}, \ldots\big). \tag{5}$$

RSI mode, d_{mode}, places improvement in an offline phase, during online use, or first offline and then online. *Update frequency*, $d_{\mathrm{frequency}}$, specifies the amount of execution or the event that triggers an update to persistent resources: a step, a specified event, a complete trajectory, or a batch. For example, a system can receive feedback after every task and revise its skills after a batch of tasks. *Experience scope*, d_{scope}, describes whether retained resources are intended for a particular task context or for reuse across task types and domains. The remaining entries of D describe the three functions within U.

Candidate generation. P uses the current system's resources together with available task material, execution records, and feedback to construct a candidate configuration $\widetilde{\mathcal{A}}_t$:

$$P\big(\mathcal{A}_t; d_{\mathrm{artifact}}, d_{\mathrm{updater}}, d_{\mathrm{topology}}\big) \longrightarrow \widetilde{\mathcal{A}}_t. \tag{6}$$

Three further entries of D describe candidate generation. *RSI artifact*, d_{artifact}, specifies which resources are modified. *Artifact updater*, d_{updater}, identifies the roles that form the update, including those that propose, revise, or curate candidate changes. *RSI topology*, d_{topology}, describes how the candidate draws on earlier versions and their relationships.

Candidates can revise memory, context, skills, or code, or update model parameters using generated learning material [13, 14, 15, 5, 2]. For example, a skill revision produces a candidate that uses the same model but contains the revised skill.

Candidate feedback. F assesses the candidate produced by P and returns the candidate paired with its feedback f_t:

$$F\big(\widetilde{\mathcal{A}}_t; d_{\mathrm{source}}, d_{\mathrm{type}}\big) \longrightarrow \big(\widetilde{\mathcal{A}}_t, f_t\big). \tag{7}$$

Assessment can involve inspecting the candidate's resources, examining execution records, or running it to obtain further results. The feedback f_t provides evidence for the retention decision. *Feedback source*, d_{source},

specifies where the feedback is obtained, while *Feedback type*, d_{type}, specifies its form, such as test outcomes, scores, or textual assessments.

Retention. R uses the current system and the pair returned by F to determine the configuration retained for the next round:

$$R\big(\mathcal{A}_t, (\widetilde{\mathcal{A}}_t, f_t); d_{\text{acceptance}}\big) \longrightarrow \mathcal{A}_{t+1}. \tag{8}$$

The *Acceptance criterion*, $d_{\text{acceptance}}$, specifies which changes to retain, using artifact validation, task results, or aggregate performance. Accepted changes are incorporated into the current system; resources that are not replaced remain available.

The settings introduced above give the complete design configuration:

$$\begin{aligned} D = \big(&d_{\text{mode}}, d_{\text{frequency}}, d_{\text{scope}}, \\ &d_{\text{artifact}}, d_{\text{updater}}, d_{\text{topology}}, \\ &d_{\text{source}}, d_{\text{type}}, d_{\text{acceptance}}\big). \end{aligned}$$

Section 3 develops these nine settings into a taxonomy of existing methods.

In the next round, $\mathcal{A}_{t+1}$ supports both execution and the construction of further candidates. A retained skill can guide a later task and serve as the starting point for another revision. Changes to the models, instructions, or code used to produce updates can also alter how subsequent candidates are constructed [5, 16, 17]. Successive applications of U therefore connect retained changes to further self-improvement.

3 A Unified Taxonomy of Agent Self-Improvement

An RSI method combines choices about what changes, who produces the changes, when they are made and reused, and how feedback guides their retention. We propose a unified taxonomy that describes these choices through nine dimensions organized into four groups. Together, the dimensions define a method design space. Describing existing methods in this space helps researchers compare their mechanisms and examine how choices along different dimensions interact. This analysis can reveal limitations of current mechanisms and inspire new research ideas for agent improvement.

3.1 Overview of the unified taxonomy

The nine design dimensions are organized into four groups, as shown in Figure 2.

1. **Update targets:** RSI artifact.
2. **Update organization:** Artifact updater and RSI topology.
3. **Update schedule and scope:** RSI mode, Update frequency, and Experience scope.
4. **Feedback and selection:** Feedback source, Feedback type, and Acceptance criterion.

Figure 3 organizes the reviewed methods by their update targets. Appendix A lists the method papers, benchmarks, and software systems covered in this survey, together with their classifications. The following sections define each dimension and illustrate its choices with representative methods.

3.2 Update targets

Update targets describe the parts of an agentic system that are modified and retained for subsequent use. We characterize these targets through the *RSI artifact* dimension.

RSI artifact. This dimension distinguishes updates to model weights (*Parametric*) from updates to resources outside those weights (*Non-parametric*). Non-parametric targets are grouped by their role in the system. *Context* supplies information to model calls; *Memory* stores information for later retrieval; *Skill* packages reusable instructions and supporting resources; and *Harness code* implements tool operations and coordinates model calls, tool execution, and control flow. *Other artifact* covers retained resources outside these categories, such as data strategies and experiment configurations. A method can update several artifacts together.

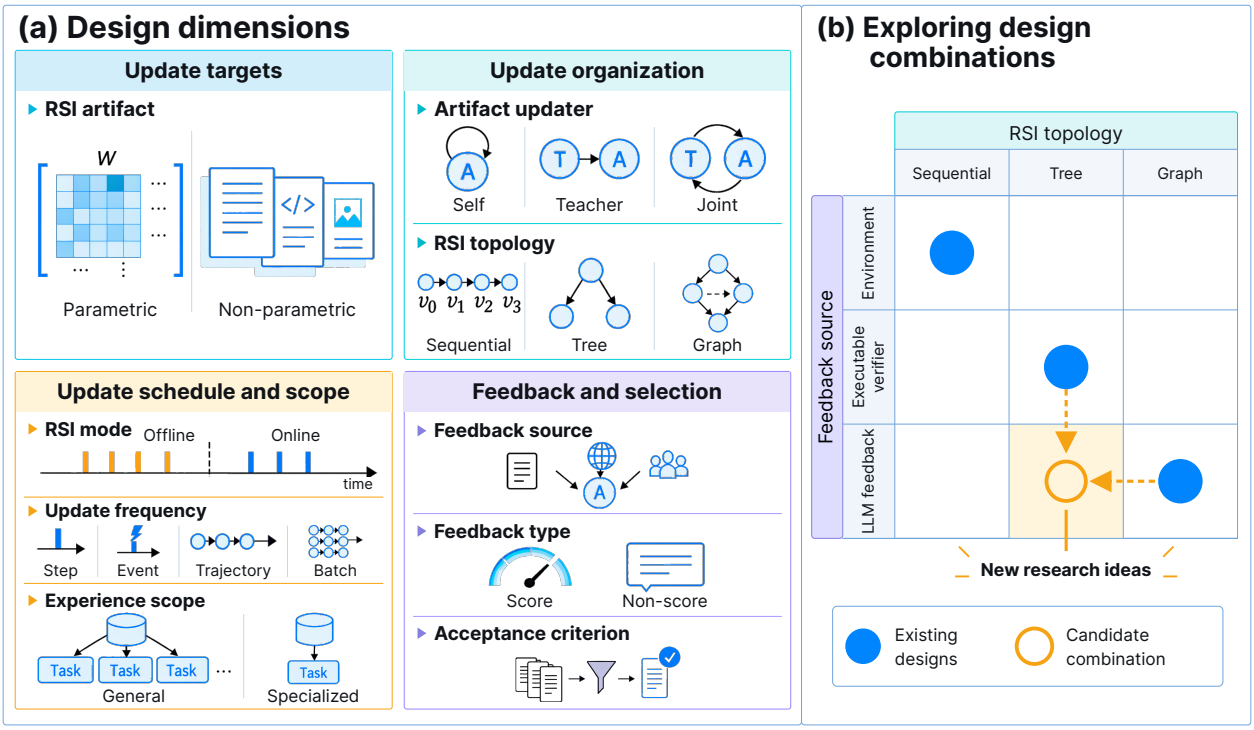


Figure 2 The unified taxonomy and its use as a design space. Left: four groups organize nine design dimensions. Right: an illustrative slice locates existing designs and a candidate direction within the same space.

Parametric methods update the language model's weights using training data and feedback. SEAL generates training material or update instructions and uses them to fine-tune the model. The updated model's task performance then serves as a reward for learning to generate more effective self-edits [2].

Context updates change the information supplied to model calls and how it is organized. They include revising task instructions and selecting, compressing, or reorganizing observations and interaction history as long-horizon tasks unfold. GEPA uses execution traces and feedback to propose revised instructions, then evaluates the candidates before retaining them [18]. ACON optimizes the guidelines used to compress observations and interaction history. It compares successful runs with full context against failed runs with compressed context, then uses these comparisons to revise the guidelines so that compression preserves information needed for subsequent decisions [19].

Memory methods retain information that can be retrieved when relevant. ReasoningBank extracts strategies from model-judged successful and failed trajectories. A-MEM organizes interaction records into structured notes and revises links among related memories [20, 21]. TTSE separates environmental facts from task-conditioned procedures, while SAGEAgent distinguishes individual decision records from reusable rules for acquiring input modalities [22, 23]. KSI maintains a shared knowledge base whose contents are discussed and refined by agents across task attempts [24].

Skill methods construct or revise reusable procedures. TRACE compares successful and failed uses of a skill to revise its contents, and Evo-Harness compiles execution records into reusable skills [25, 26]. SkillPivot uses a teacher's continuation from the same trajectory prefix as a failed attempt to guide a localized skill edit [27].

Harness code methods change executable behavior. DGM edits the coding agent's program and retains evaluated versions. Hyperagents allows both the task agent and the meta-agent that proposes improvements to be modified [5, 17]. The code target can therefore include the implementation of task execution or of the update procedure.

Several methods span multiple artifact categories. SkillAdam revises a skill while retaining a record of earlier editing attempts, and SkillSmith revises skills together with their executable tools [4, 28]. AlgoEvo stores

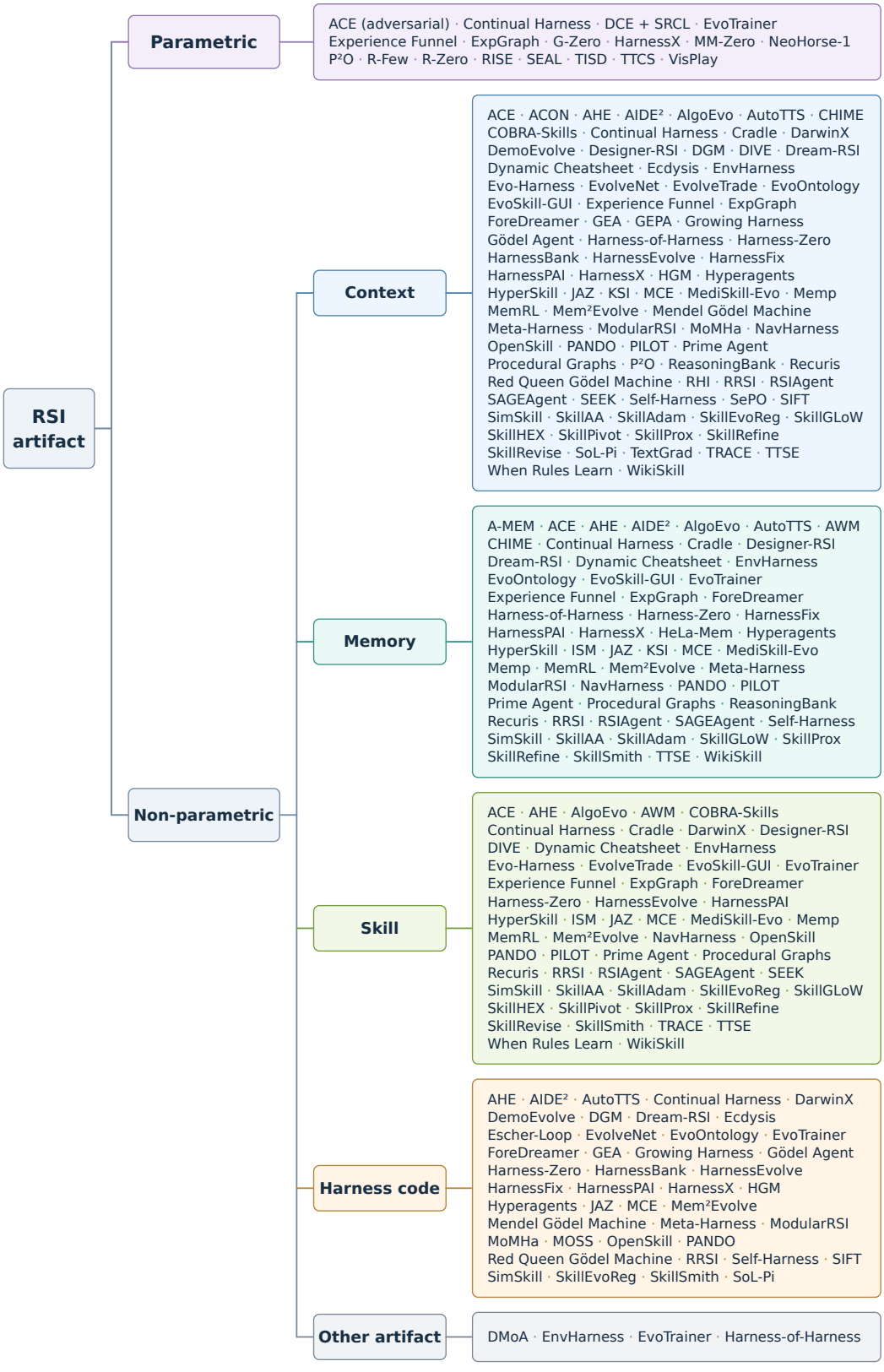


Figure 3 Methods grouped by RSI artifact. Papers with multiple labels appear on multiple branches.

evaluated modifications in persistent cards and consolidates recurring patterns into design skills, linking Memory with Skill [29]. RRSI permits coordinated changes to harness code, context, memory, and skills around a frozen model [10].

3.3 Update organization

Update organization describes how the system produces successive versions of itself. It combines two dimensions: *Artifact updater* identifies who forms a change, and *RSI topology* describes how a new version builds on earlier versions.

Artifact updater. This dimension distinguishes *Self*, *Teacher*, and *Joint* updates. Self denotes changes produced by the agent being improved. Teacher denotes changes produced by a separate updating role using the agent's execution records and feedback. Joint combines contributions from the agent and a teacher: the agent proposes or revises a change, and the teacher further refines, curates, or selects it.

SEAL provides a Self example: the model generates material used to update its own parameters [2]. Recuris assigns skill-memory revision to a meta-agent, while GEPA uses a reflective optimizer to revise the evaluated system's instructions; both illustrate Teacher updates [30, 18]. DGM illustrates Joint updating: a diagnosis guides a self-modification attempt by the selected coding agent, and evaluation determines whether the resulting version enters the archive [5].

RSI topology. This dimension describes dependencies between versions through *Sequential*, *Tree*, and *Graph* topologies. Sequential improvement repeatedly updates the current version to form a single chain. A Tree allows archived versions to branch, with each child derived from one parent. A Graph allows a new version to draw on retained state, changes, or outcomes from several historical versions or lineages.

A-MEM illustrates Sequential updates by incorporating new notes and revising the current memory store [21]. DGM selects an archived agent as a parent and modifies its code to produce a child; repeatedly branching from archived agents forms a Tree [5]. HGM also maintains a branching archive and uses descendant performance to guide the choice of promising branches [31].

Mendel Gödel Machine illustrates Graph dependencies. It compares the target agent's execution trace with that of an agent from another lineage on the same task, then uses the comparison to guide code changes [32]. GEPA's Merge variant draws on complementary changes from different candidates [18]. In both cases, a new version can inherit useful information from more than one lineage.

3.4 Update schedule and scope

Update schedule and scope describes when changes are made and which tasks they are intended to support. Three dimensions specify these choices: *RSI mode* places improvement relative to system use, *Update frequency* records when individual updates are triggered, and *Experience scope* describes the intended range of reuse.

RSI mode. This dimension distinguishes *Offline*, *Online*, and *Offline* $\rightarrow$ *Online* improvement. Offline improvement occurs in a separate training or search phase before evaluation or deployment. Online improvement occurs during use, so later tasks can use the newly retained state. Offline $\rightarrow$ Online first builds a resource offline and then continues to update it during use.

DGM and SkillAdam illustrate Offline improvement by searching for code or skill revisions before evaluating the resulting system [5, 4]. A-MEM and ReasoningBank illustrate Online improvement by incorporating new records or distilled strategies as tasks are processed [21, 20]. Agent Workflow Memory studies both offline workflow construction and continued online induction, including the use of offline workflows to initialize online improvement [33]. ACE separately studies offline playbook construction and online updating [15].

Update frequency. This dimension describes the trigger or unit of an individual update through *Step*, *Event*, *Trajectory*, and *Batch*. A Step update follows one action–observation step. An Event update is triggered by a specified condition, such as detecting a failure or receiving new information. A Trajectory update follows a complete task execution, while a Batch update uses a group of trajectories or candidates.

ReasoningBank extracts memories from completed trajectories, whereas SkillAdam uses batch-level results to guide skill revisions [20, 4]. PILOT illustrates how different update frequencies can coexist. A supervisor

can redirect an active worker and record reusable skills or memory during execution. After a group of runs finishes, changes from successful runs are merged into the shared state used by the next iteration [34]. These operations correspond to Event and Batch updates, respectively.

Experience scope. This dimension describes the tasks for which an updated artifact is intended to be reused. *General* artifacts support reuse across task types or domains. *Specialized* artifacts serve a particular instance, topic, or task family. A system can retain resources with different scopes.

DGM develops a reusable coding harness and studies its transfer between SWE-bench and Polyglot, illustrating General scope [5]. SkillAdam refines skill documents for particular task settings, illustrating Specialized scope [4]. AlgoEvo reuses experience both within and across tasks: search records guide further attempts on the current algorithm-design task, while skills distilled from multiple tasks guide subsequent algorithm-design tasks [29].

3.5 Feedback and selection

Feedback and selection describes how the system obtains guidance for updates and decides which changes to retain. We separate three dimensions: *Feedback source* identifies where the guidance comes from, *Feedback type* describes its form, and *Acceptance criterion* specifies the basis for retaining a change.

Feedback source. This dimension groups information by its origin into six categories. A *Train/dev set* provides task examples and reference outputs; a *Benchmark* supplies results from a standardized evaluation suite. The *Environment* provides observations and action outcomes, while an *Executable verifier* produces test or validation results. *LLM feedback* supplies model-generated evaluations or suggestions. A *Human* can provide corrections, preferences, or guidance. An update can use several sources.

DGM uses coding-benchmark results and executable tests to assess revised agents [5]. SIFT adds model-generated pairwise comparisons between candidate patches to guide parent selection and prioritize execution-based evaluation [35]. SkillHEX combines model-generated failure hypotheses with executable checks: it tests the hypotheses on cached outputs from skill versions and uses the results to guide later revisions [36].

Feedback type. This dimension describes the form of feedback, distinguishing *Score* and *Non-score* information. Score feedback summarizes an outcome or assessment as a *Binary* result, such as success or failure, or a *Non-binary* numerical or categorical evaluation. Non-score feedback supplies content for revising the system: *Ground truth* provides a reference answer or behavior, *LLM-as-a-judge* provides a textual assessment, and *Other* covers observations, execution records, and diagnostics. The type is determined by the returned information: a numerical model rating is a Score, while its explanation is Non-score feedback.

GEPA uses execution scores to compare candidates and detailed execution feedback to guide revisions [18]. ReasoningBank judges whether a trajectory succeeded and extracts lessons from both successful and failed attempts [20]. ACE's reflector analyzes execution and produces guidance for playbook updates [15]. These methods use both compact assessments and information about the behavior that produced an outcome.

Acceptance criterion. This dimension describes how a system decides whether to retain a proposed change. *Artifact validation* checks whether the modified resource meets requirements such as successful compilation or interface compatibility. *Instance result* uses performance on an individual task. *Benchmark score* uses aggregate performance on a benchmark. *Combined metrics* considers several measures, such as performance, cost, and stability. *No validation* denotes direct retention without a separate acceptance check.

GEPA evaluates candidate instructions before adding successful revisions to its search population [18]. Designer-RSI compares candidate and reference skills on matched replays with the same upstream state and retrieved assets. It admits a change when at least one replayed case improves and no regression is detected on the others [37]. RRSI considers performance gains together with execution cost and evaluation noise, illustrating Combined metrics [10]. ACE and A-MEM illustrate No validation by incorporating curated playbook edits or newly constructed memory records without a separate performance-based acceptance test [15, 21].

3.6 Ideas from the unified taxonomy

This section illustrates how the unified taxonomy can help generate new RSI research ideas. Starting from existing methods, we examine the relationship between relevant dimensions, identify a limitation of their current designs, and use it to motivate a research direction.

We begin with *Harness code* under the *RSI artifact* dimension. Code updates can change task execution or the information recorded for failure diagnosis. HarnessFix can modify model-input logging, context snapshots, and state-change records [38]. EvoTrainer also revises its diagnostic metrics and analysis procedures when they do not provide enough information to guide further changes [39].

We illustrate this with an agent that can modify its own harness. During task execution, it uses a web research tool to find papers published within a given date range, but its requests repeatedly fail. During self-improvement, the same agent examines its logs, discovers that the request parameters are missing, and edits its logging code to record them. When it runs the search task again, the new logs reveal a date format that the tool does not accept. In the next round of self-improvement, it fixes the code that prepares the dates. The first edit does not resolve the search failure, but supplies the evidence needed for the later repair.

Whether the first change is retained, however, depends on the *Acceptance criterion*. Ecdysis, for example, retains a candidate harness only when its overall training score exceeds that of the current version [40]. Under such a rule, a logging change that leaves the score unchanged would be rejected before it could support a later repair. This reveals a mismatch between the two dimensions: code updates can improve failure diagnosis, while an acceptance rule based on immediate task scores may overlook that contribution.

This mismatch motivates a research direction: acceptance criteria that account for a change's contribution to subsequent self-improvement. For harness code, the criteria could consider both current task performance and whether an edit helps the agent diagnose and repair failures in later updates. The taxonomy guides this idea by exposing a gap between what the modified resource can contribute and what the current acceptance rule measures.

4 Insights from Traditional Deep Learning

Traditional deep learning can inform the design of RSI methods. We first examine how existing work uses learning and optimization mechanisms for agent self-improvement. We then explore how further developments of these mechanisms can inspire new research ideas, extending the discussion to broader methods in machine learning and evolutionary computation.

4.1 Connections in Existing RSI Methods

Existing RSI methods draw on traditional deep learning to guide how agents learn from feedback and modify themselves. This section examines the problems these learning mechanisms were developed to address and how representative RSI methods apply them to self-improvement. Figure 4 summarizes these connections.

4.1.1 Optimization

Optimizers in traditional deep learning determine the direction and magnitude of parameter updates. Adam, for example, uses information accumulated from current and past gradients to compute its parameter updates [41]. Iterative skill revision in RSI faces a similar problem: deciding what to modify and how much to change at each iteration.

SkillAdam draws on this optimizer design to guide the revision of agent skills [4]. It records identified problems, attempted fixes, and their outcomes, allowing later revisions to build on earlier attempts. It also compares task performance before and after a revision to determine how much of the skill may be changed next. When the gains vary substantially across evaluation cases, it reduces the amount of editing allowed in the next iteration; more consistent gains permit broader changes. These mechanisms use the history and results of previous revisions to guide both what to change and how much to change.

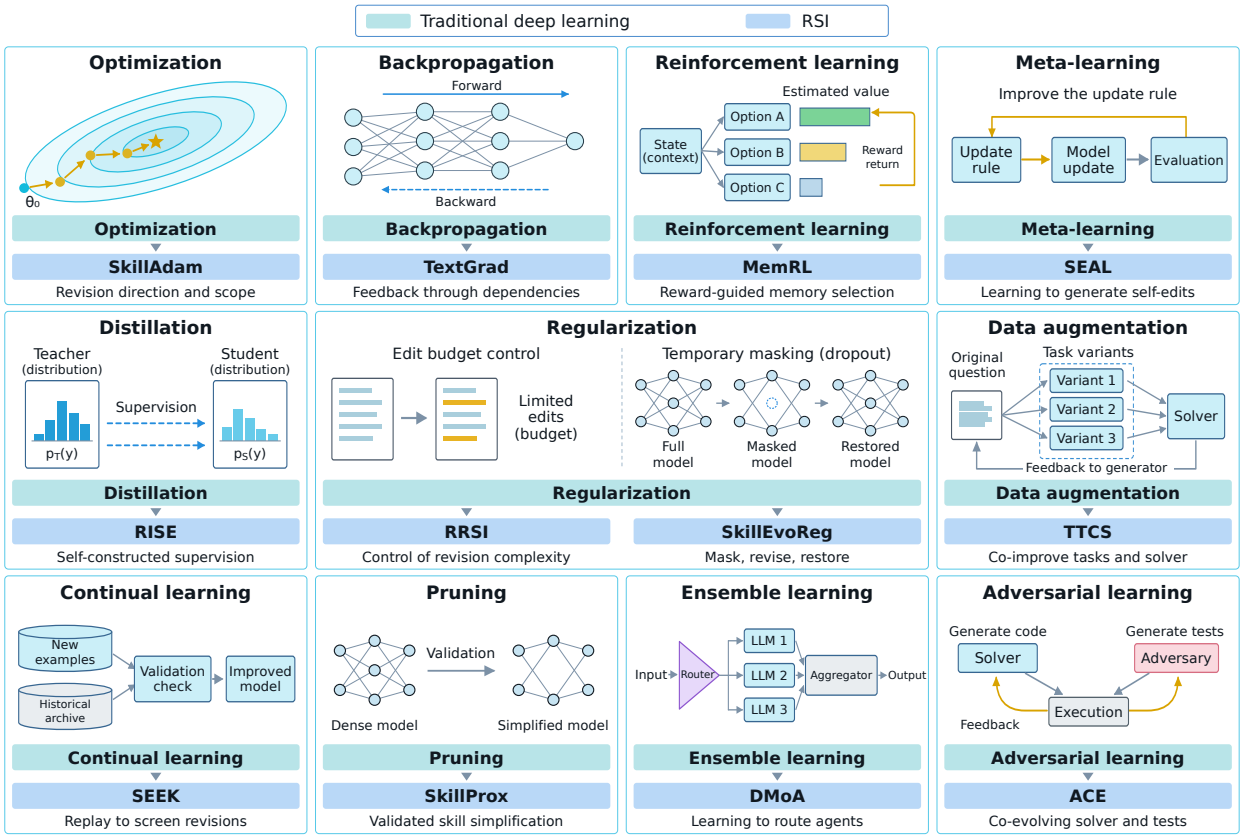


Figure 4 Connections between traditional deep learning and RSI, illustrated by representative methods. Each panel links a learning mechanism to its use in RSI; the regularization panel includes both complexity control and temporary masking. Network sketches illustrate the learning principles; SkillEvoReg masks skill instructions, while SkillProx simplifies skill contents.

4.1.2 Backpropagation

Training a neural network requires using errors in its predictions to adjust parameters across its layers. Backpropagation works backward through the computation graph to calculate how the loss changes with each parameter, providing gradients for parameter updates [42]. RSI systems face a related problem: using feedback on a task result to guide changes to individual components.

TextGrad draws on this idea to propagate textual feedback through a system [9]. It represents operations and their input–output relationships as a computation graph, then passes feedback on the final result backward through that graph. At each step, a language model uses the local inputs, outputs, and feedback to generate revision suggestions for upstream variables. An optimizer then uses these suggestions to revise selected variables, such as instructions and code. For RSI, this suggests a way to turn task-level feedback into specific changes by following the steps that produced the result.

4.1.3 Reinforcement Learning

Reinforcement learning uses rewards from interaction to improve decisions. Value-based methods estimate the expected return of available choices and use these estimates to guide selection [43]. This principle can also guide memory retrieval: an agent can use task outcomes to learn which memories are useful and adjust its selection on subsequent tasks.

MemRL applies value estimation to this selection process [3]. It first retrieves a pool of memories by semantic similarity, then selects from this pool using both relevance and learned utility values. After the agent executes the task, the resulting reward updates the utility values of the memories supplied to the agent. Memories

associated with higher rewards are consequently more likely to be selected for later tasks. With the language model's parameters fixed, feedback improves how the agent selects from its accumulated records.

4.1.4 Meta-Learning

Meta-learning uses performance after learning to improve how learning is carried out. For example, it can optimize a model's initialization or learn a parameter-update rule by evaluating how well the model performs after training [44, 45]. In RSI, this principle can guide the production of training data and update instructions: their quality can be assessed through the learning they produce.

SEAL implements this idea by training a model to generate its own self-edits [2]. Given new information or a task, the model produces training materials or instructions for updating its parameters. The proposed update is applied through fine-tuning, and the updated model is evaluated on the relevant task. Its performance then serves as a reward for reinforcement learning of the self-edit generator. This creates two connected learning processes: one updates the model using generated materials, and the other improves how those materials and update instructions are produced.

4.1.5 Distillation

Knowledge distillation transfers behavior from a teacher model to a student by training the student to match the teacher's output probabilities [46]. These probabilities provide guidance about alternative outputs beyond a single target answer. A self-improving system can use this mechanism while also constructing and updating the source of its supervision.

RISE combines distillation with reinforcement learning using verifiable rewards [47]. It compares the model before and after a reinforcement learning update, then constructs a teacher by extending the observed change farther in the same direction. This extrapolation can operate on model weights or output log-probabilities. The student is trained to match the resulting teacher's token probabilities. As reinforcement learning and distillation alternate, each new training update provides the basis for another teacher. RISE thus incorporates teacher construction into the improvement process, using the model's own training trajectory to produce supervision for subsequent learning.

4.1.6 Regularization

Regularization aims to reduce overfitting and improve generalization beyond the training data. Common approaches constrain model complexity or introduce perturbations during learning. L_0 regularization penalizes the number of nonzero parameters to encourage sparsity [48]. L_1 regularization encourages sparsity by penalizing the sum of absolute parameter values, while L_2 regularization discourages large weights by penalizing the sum of squared parameter values [49]. Dropout introduces perturbations by temporarily masking units during training, reducing reliance on particular combinations of features [50]. RSI faces a similar difficulty when repeated harness or skill revisions fit the tasks used for improvement but transfer poorly to new tasks.

RRSI applies complexity control to harness evolution [10]. It gradually reduces the number of independent edits allowed in each proposal, limiting how many changes can be introduced in response to a round of feedback. Components that repeatedly fail to yield gains are marked for removal. Candidate selection also requires increased execution cost to be justified by improved task performance.

SkillEvoReg adapts dropout to skill revision [51]. It randomly masks instructions or procedural blocks in the current skill. The agent executes a training task with this partial skill, and the updater uses the resulting execution to propose a revision. The system then restores the masked content and integrates the proposed changes into the complete skill. The aim is to reduce the system's reliance on particular existing instructions when revising a skill.

4.1.7 Data Augmentation

Data augmentation expands training data by constructing variants of existing examples. In image classification, for example, crops and horizontal reflections provide additional views of labeled images [52]. For a self-improving agent, task variants can similarly provide further practice on the reasoning or operations required by the problems it encounters.

TTCS uses a question synthesizer to generate variants of test questions [53]. It changes problem objects, settings, or constraints while aiming to preserve the underlying reasoning structure. The solver trains on both original and generated questions, using agreement among sampled answers as a learning signal. Solver feedback also updates the synthesizer. The system thus learns from the additional data while improving how those data are generated.

4.1.8 Continual Learning

Continual learning concerns learning from successive tasks or data while retaining earlier capabilities. Rehearsal reuses past examples during later learning; iCaRL combines this with distillation to preserve previously learned classes [54]. Agent self-improvement raises the same retention problem when a new skill revision disrupts tasks that the agent could previously complete.

SEEK uses historical replay to screen such revisions in an industrial search evaluator [55]. Reviewed errors guide changes to the operational instructions in a skill bank. Each candidate is then checked on emerging cases and a fixed set of historical examples. Acceptance requires improvement on the new cases while limiting regression on the historical set. The retained examples serve as a check on successive skill updates, extending the role of past data from parameter training to the acceptance of reusable instructions.

4.1.9 Pruning

Pruning removes network weights or structures that contribute little to the desired behavior, often followed by further training [56]. For agents, repeated revisions can similarly accumulate instructions and resources whose contribution no longer justifies keeping them.

SkillProx implements a related simplification step after skill revision [57]. It measures the utility of skill sections and reference groups by removing them and rerunning validation cases. Low-utility units become candidates for deletion, demotion, or consolidation. Each proposed simplification must reduce complexity, preserve structural validity, and pass performance checks. The resulting skill becomes more compact through measured changes to its contents. The method also draws on proximal optimization by separating performance-oriented revision from subsequent complexity reduction.

4.1.10 Ensemble Learning

Ensemble learning combines predictions from multiple learners to use their complementary strengths [58]. In agentic systems, this connection concerns how several agents contribute to a solution and how their participation is learned.

DMoA learns a router that selects agents at successive reasoning steps, combines their responses into the next step's context, and uses a summarizer to produce the final answer [59]. Its test-time training setting first runs all candidate agents on an initial set of queries and collects predictive entropy as supervision. It then updates the router and uses the learned selection on subsequent queries. This protocol illustrates how a system can learn to coordinate an ensemble from its own outputs while keeping the underlying language models fixed.

4.1.11 Adversarial Learning

Adversarial learning trains interacting models with competing objectives. In generative adversarial networks (GANs), a discriminator learns to distinguish real data from generated samples, while a generator learns to produce samples that the discriminator classifies as real [60]. As the two models are updated, each learns against a changing opponent. In RSI, a related interaction can couple task solving with the search for failures in current solutions.

The code-generation framework ACE illustrates this interaction through a solver and an adversary [61]. The solver generates candidate programs, while the adversary generates test inputs intended to trigger execution failures. The system evaluates the programs using both existing correctness tests and the generated inputs, then fine-tunes the solver on selected programs. Execution results also provide preferences for training the adversary: tests are favored when they cause some candidates to fail while others still execute successfully. Each new round uses the updated solver and adversary, allowing failure discovery and code generation to improve through continued interaction.

4.2 From Established Methods to New RSI Ideas

This section explores how further developments of learning methods can inspire new RSI research ideas. We begin with deeper developments in traditional deep learning, then broaden the discussion to other machine learning methods and evolutionary algorithms. Figure 5 places these sources within a wider landscape of connections to RSI.

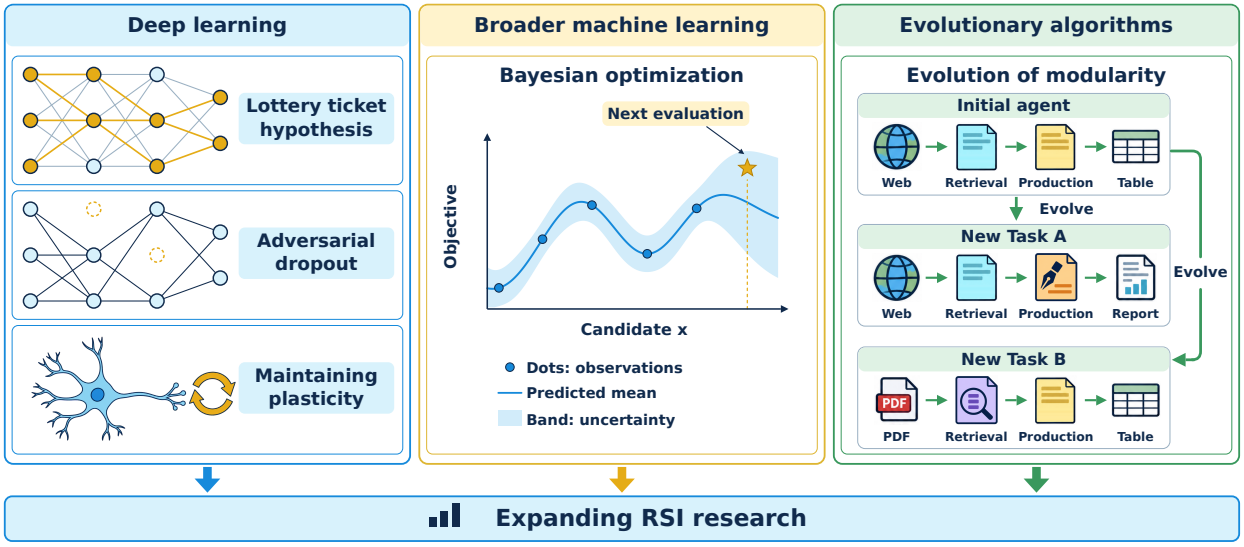


Figure 5 Broadening the sources of ideas for RSI. Left: the lottery ticket hypothesis, adversarial dropout, and maintaining plasticity in deep learning. Center: Bayesian optimization. Right: evolution of modularity, illustrated by revising production or retrieval modules to adapt to new tasks. Bottom arrows indicate how these sources can inspire RSI research directions.

4.2.1 Further Developments in Traditional Deep Learning

We first deepen the connections established in Section 4.1, considering more specific mechanisms from traditional deep learning.

The Lottery Ticket Hypothesis. SkillProx uses pruning to simplify skills while preserving their task performance, as discussed in Section 4.1.9. The lottery ticket hypothesis takes pruning a step further: a large network can contain a small subnetwork that, when trained from its original initial weights, reaches comparable accuracy in a similar number of iterations [62].

For RSI, the lottery ticket hypothesis suggests examining which resources an agent needs at the start of self-improvement. A system with an extensive collection of skills and code might need only a subset of its early resources to develop comparable capabilities through subsequent revisions. If such a starting point exists, it could enable agents to improve effectively with fewer resources.

Adversarial Dropout. Dropout randomly masks units during training. Adversarial dropout develops this idea by seeking bounded changes to the mask that most increase prediction error, then training the network to cope with them [63]. SkillEvoReg already applies random masking to agent skills during execution and

revision [51]. This connection suggests using targeted masking to expose strong dependencies in skills and encourage alternative procedures.

For example, a masking agent could identify the skill content on which a web research agent strongly depends. If hiding a template tailored to one website causes extraction to fail, it could use that disruption to encourage the system to explore other extraction procedures, such as locating information through page headings and nearby text. As the skills improve, the masking agent could seek further masks that expose weaknesses, providing new challenges for subsequent self-improvement.

Maintaining Plasticity. Section 4.1.8 describes how past examples help retain earlier capabilities. Continual learning also requires plasticity: the ability to learn new behavior through further training. This ability can decline as a network learns successive tasks. Continual backpropagation addresses this problem by selectively reinitializing a small fraction of low-utility hidden units during training, allowing them to learn new features [64].

An agent that improves itself through parameter updates can use its own verified task solutions as training data. However, prolonged training may gradually reduce the model's ability to learn new ways of solving tasks. Drawing on continual backpropagation, the system could reinitialize a small fraction of low-utility neurons during training, then resume training so these neurons can learn new features. This could help the agent maintain its ability to learn from new tasks across successive rounds of self-improvement.

4.2.2 Insights from Broader Machine Learning

Broader machine learning methods can also inspire new research ideas for RSI. We illustrate this with a concrete example below.

Bayesian Optimization. Bayesian optimization uses a probabilistic model of previous evaluations to guide the choice of subsequent evaluations. Freeze-Thaw Bayesian Optimization extends this approach to partially completed training runs [65]. It uses a Gaussian process to predict future performance from observed learning curves and estimate the uncertainty of these predictions. This information guides decisions to continue a training run, pause it, or resume a previously paused run.

An RSI system exploring several branches of self-improvement could use this mechanism to decide which branches to develop further. Following Freeze-Thaw, a lightweight Gaussian process predictor could use the performance recorded after successive self-modifications to estimate what additional rounds might achieve. For example, one branch of a web research agent might score well but make little recent progress, while another starts lower and continues to improve. Predicted performance and its uncertainty could guide which branch to continue, pause, or resume. Paused branches would retain the agent state and revision history needed to continue their self-improvement. As further revisions are evaluated, their results would update the predictor and guide the allocation of subsequent improvement rounds.

4.2.3 Insights from Evolutionary Algorithms

Evolutionary algorithms can likewise inspire new research ideas for RSI. We illustrate this with a concrete example below.

Evolution of Modularity. Changing one part of a skill can disturb another part that already works. Separating procedures that can change independently could make later revisions easier. ModularRSI, for example, improves predefined harness modules separately [66]. Evolutionary experiments suggest that useful divisions can also emerge as task requirements change. Kashtan and Alon [67] repeatedly switched between goals made from different combinations of shared subgoals. The networks evolved modules that could be reused when the goal changed.

For RSI, consider a research agent that initially uses one skill to retrieve webpages and produce a comparison table. Later tasks sometimes require a written report from the same webpages, and sometimes require the same table from PDF documents. During self-improvement, the updater could identify which procedures need changing in each case: a new source requires different retrieval steps, while a new output format requires different reporting steps. It could then separate these procedures into reusable skills. The agent could subsequently learn to read a new document type while retaining its report-writing skill, or improve report writing while retaining its retrieval skill.

5 Evaluation and Benchmarks

RSI evaluation examines how self-improvement changes an agent's ability to solve tasks. A benchmark defines a *problem setup* through the tasks it provides, the feedback available to the system, the updates it allows, and the way it assesses the resulting changes. These choices determine which aspects of self-improvement the evaluation measures.

A central distinction is how improvement is combined with testing. We organize benchmarks and evaluation settings into *Online*, *Offline*, and *Offline* → *Online* according to whether the system improves during evaluation, before evaluation, or in both phases. Within each setting, we compare task organization, available feedback, and evaluation goals. Table 1 compares representative studies, and Appendix A provides the full benchmark catalogue.

Table 1 Comparison of representative RSI benchmarks and evaluation settings.

Study	Domain	Evaluation setup	Feedback for improvement	Evaluation focus
Online				
Evo-Memory [68]	Reasoning, coding, and interactive tasks.	Streams with varying task difficulty and order.	Correctness signals after execution.	Performance gains from using earlier task records.
SkillFlow [69]	Finance, healthcare, documents.	Related tasks in increasing difficulty.	Execution traces and verifier feedback.	Skill acquisition, repair, and reuse.
AgentStream [70]	Web research, coding, and tool use.	Isolated, sequential, or interleaved domains.	Execution results and self-evaluation.	Transfer and interference across domains.
PATH-Bench [71]	Code generation and tool use.	Probes repeated after helpful or interfering examples.	Execution records between probes.	Transfer and forgetting.
Offline				
MemoryBench [72]	QA and writing: general, legal, academic.	Separate training and test examples.	Simulated user comments and actions.	Generalization to unseen examples.
GDPevo [73]	Enterprise workflows.	Rules learned separately are combined in held-out workflows.	Reference answers, grader scores, or observations.	Transfer to new combinations of learned rules.
HarnessOpt-Bench [74]	Document QA, search, and terminal tasks.	Select a candidate harness, then test it on held-out examples.	Development results and traces, aggregate validation scores.	Held-out gain over the seed harness.
S3Gym [75]	Text-based games.	Exploration followed by held-out evaluation.	Environment observations and self-judgments.	Self-judgment accuracy and held-out gains.
Offline → Online				
AWM [33]	Web navigation on Mind2Web.	Training-derived workflows followed by updates during testing.	Demonstrations and model success judgments.	Transfer to new tasks, websites, and domains.
Mem^{2}Evolve [76]	QA, math, planning, and interactive tasks.	Memory from prior examples, updated during testing.	Model success judgments and critiques.	Cross-benchmark transfer and continued learning.

5.1 Online evaluation

Online evaluation allows a system to update itself during task execution and use retained changes in subsequent work. It examines whether the system improves through continued use and how its performance changes as new tasks arrive. The relationships among tasks and the feedback they provide shape the opportunities for

improvement.

To examine reuse across tasks, Evo-Memory organizes reasoning, question-answering, and interactive tasks into streams [68]. The system retrieves memory before solving a task and updates it after the interaction, using records of its execution and correctness feedback. Evaluating later tasks with the accumulated memory measures how well the system uses information from earlier interactions. The benchmark also varies task difficulty order and feedback to examine how these conditions affect learning.

Tasks that share a procedure provide a more specific test of reuse. SkillFlow constructs task families around a common workflow [69]. Agents start without skills, solve the tasks sequentially, and use execution traces and feedback based on scoring rubrics to create or repair skill documents. The updated library remains available for later tasks in the family. ContinualSkillBench arranges related tasks by difficulty and opportunities for skill reuse [77]. It compares independent task execution, accumulated context, and explicit skill maintenance. These settings examine whether what is learned from one task helps solve related tasks, and how the form of retained information affects that transfer.

Changing the task sequence makes transfer and retention observable. AgentStream organizes agent tasks into isolated, sequential, and interleaved domain settings [70]. This allows comparisons between learning within one domain and carrying updates across domains. PATH-Bench places recurring probe tasks among histories designed to help or interfere with them [71]. Revisiting the probes reveals whether later updates preserve earlier capabilities or cause forgetting. The sequence therefore forms part of the problem being evaluated: performance on a task depends on what the agent encountered before it.

Online evaluation can also track successive attempts at an ongoing research objective. In RSIBench-Data, a research agent repeatedly revises training-data strategies for a target model [78]. A fixed training and evaluation pipeline returns results after each experiment, which inform the next revision. This setting examines how an agent uses experimental feedback to improve its research decisions over a sequence of attempts.

5.2 Offline evaluation

Offline evaluation separates improvement from testing. The system first learns or searches on training tasks, then solves held-out tasks with further self-improvement updates disabled. The evaluation asks whether the changes acquired during improvement generalize. The relationship between training and test tasks specifies the kind of transfer being examined.

MemoryBench tests transfer from feedback on training tasks to separate test tasks [72]. A simulated user provides feedback on the system's training responses, and a separate evaluation process scores its test responses. The tasks span several domains, languages, and input–output formats. This setup examines whether feedback can support learning that is useful beyond the examples that produced it.

Some benchmarks explicitly construct the connection between learning and test tasks. GDPevo distributes business rules across training tasks and recombines them in held-out enterprise workflows [73]. The system must apply rules it has learned in combinations it has not practiced. EvoAgentBench links training and test tasks through shared procedural requirements [79]. These designs make the expected transfer concrete by specifying which procedures learned during improvement should help with a later test task.

Offline evaluation also tests whether harness revisions improve performance on unseen tasks. HarnessOpt-Bench supplies an initial harness and feedback for revising it, then compares the selected version with the initial one on separate test tasks [74]. HarnessDev evaluates both harness construction and subsequent improvement, testing the resulting programs on unseen tasks after their code is frozen [80]. To reveal the effects of harness revision, Evo-Bench prioritizes tasks whose scores reflect differences in harness quality, then divides them into separate sets for improvement and final testing [81].

Offline evaluation also examines learning from exploration. In S3Gym, an agent explores text-based games and judges its own behavior before stricter held-out assessment. It can carry information forward through interaction history, summary memory, or parameter updates [75]. This setup tests how information acquired through exploration supports later task solving.

5.3 Offline-to-online evaluation

Offline-to-online evaluation begins with resources acquired during an offline phase and allows further updates during evaluation. It examines how initial preparation supports subsequent tasks and how the system continues learning from new interactions. Both the source of the initial resources and the relationship between earlier and later tasks are therefore part of the problem setup.

Agent Workflow Memory (AWM) provides an example through its combined evaluation on Mind2Web [33]. It first induces reusable web workflows from training examples, then adds workflows learned while processing test tasks. The cross-task, cross-website, and cross-domain splits vary how far those tasks depart from the training distribution. Comparing the combined setting with offline-only and online-only settings examines how previously learned workflows interact with those acquired during continued use.

The memory-initialization experiments in Mem^{2}Evolve study a related setup [76]. The system first builds memory from a subset of benchmark tasks, then continues to use and update memory while processing the remaining tasks. Another setting initializes memory from GAIA before moving to other benchmarks. These arrangements examine how resources learned within the same task family or elsewhere support continued improvement.

5.4 Toward unified evaluation

Evaluation settings differ across studies. Online, offline, and offline-to-online evaluation allow self-improvement at different stages and assess different aspects of improvement. Even within the same mode, studies can differ in the benchmarks they use, the order of evaluation examples, and the feedback they provide. For example, what a system learns from earlier examples can help it solve later ones, while reference answers, test results, and model-generated assessments provide different information for revision. Comparing the gains reported by different methods therefore requires considering these evaluation conditions.

Establishing shared benchmarks and evaluation protocols is important for RSI research. Within each evaluation mode, common task organization, feedback conditions, and scoring rules would make comparisons more informative. Reporting task performance together with the gain over the same system before improvement would further clarify the contribution of self-improvement. This common basis would help identify designs that support generalization, continued improvement, and retention of acquired capabilities.

6 Conclusion

This survey systematically reviews existing RSI methods through a unified taxonomy and examines how some of these methods draw on traditional deep learning. The taxonomy places existing methods in a shared high-dimensional design space, providing a basis for understanding and comparing them. Examining individual dimensions and their interactions can reveal limitations of current mechanisms and suggest new design directions. Through concrete examples, we explore how more specific theories and mechanisms within established learning methods can inspire new RSI research ideas, and extend the search for inspiration beyond traditional deep learning.

References

[1] Irving John Good. Speculations Concerning the First Ultraintelligent Machine. *Adv. Comput.*, 6:31–88, 1965. doi: 10.1016/S0065-2458(08)60418-0. URL https://doi.org/10.1016/S0065-2458(08)60418-0.

[2] Adam Zweiger, Jyothish Pari, Han Guo, Yoon Kim, and Pulkit Agrawal. Self-Adapting Language Models. In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*, 2025. URL http://papers.nips.cc/paper_files/paper/2025/hash/6b41e04c41726e2a60e456d0a2b961ab-Abstract-Conference.html.

[3] Shengtao Zhang, Jiaqian Wang, Ruiwen Zhou, Junwei Liao, Yuchen Feng, Weinan Zhang, Ying Wen, Zhiyu Li, Feiyu Xiong, Yutao Qi, Bo Tang, and Muning Wen. MemRL: Self-Evolving Agents via Runtime Reinforcement Learning on Episodic Memory. *CoRR*, abs/2601.03192, 2026. doi: 10.48550/ARXIV.2601.03192. URL https://doi.org/10.48550/ARXIV.2601.03192.

[4] Gaoyuan Li, Meihao Fan, Yizhe Liu, Shaolei Zhang, Ju Fan, Siyi Wang, Jiaheng Hou, Xudong Weng, Honghan Tian, and Zang Li. SkillAdam: Stable and Efficient Skill Evolution for Agents. arXiv preprint arXiv:2609.08944, 2026. URL https://arxiv.org/abs/2609.08944.

[5] Jenny Zhang, Shengran Hu, Cong Lu, Robert T. Lange, and Jeff Clune. Darwin Godel Machine: Open-Ended Evolution of Self-Improving Agents. *CoRR*, abs/2505.22954, 2025. doi: 10.48550/ARXIV.2505.22954. URL https://doi.org/10.48550/ARXIV.2505.22954.

[6] Yi Duan, Ying Liu, Zirui Tang, Haodong Chen, Jun Zhou, Yumou Liu, Bangrui Xu, Yukai Wu, Sidi Chen, Yuhan Zhou, Haoyu Wang, Xiaoyou Yu, Shaokun Han, Xuzhou Zhu, Le Zhou, Bolin Lu, Wei Zhou, Jiachen Liu, Nuozhou Fang, Jiaxin Tian, Ruoyu Chen, Yuxuan Li, Kai Zuo, Kaiyan Zhang, Qianyu Yang, Zijie Wang, Jiantao Qiu, Conghui He, Guoliang Li, Bowen Zhou, Zhiyuan Liu, Zhoufutu Wen, Jihua Kang, Xuanhe Zhou, and Fan Wu. The Last AI Built by Humans: Toward Genuine Recursive Self-Improvement. arXiv preprint arXiv:2609.11873, 2026. URL https://arxiv.org/abs/2609.11873.

[7] Mingguang Chen, Licheng Wang, and Bo Qu. Recursive Self-Improvement in AI: From Bounded Self-Refinement to Autonomous Research Loops. *CoRR*, abs/2607.07663, 2026. doi: 10.48550/ARXIV.2607.07663. URL https://doi.org/10.48550/ARXIV.2607.07663.

[8] Hanjing Li, Qiguang Chen, Chenyuan Zhang, Qionglin Qiu, Fanqing Meng, Mengkang Hu, Libo Qin, and Min Zhang. Towards AI That Improves Itself: A Survey of Recursive Self-Improvement. Author-hosted manuscript, 2026. URL https://lee1003-lee.github.io/Awesome-RSI-Research/assets/rsi-survey.pdf.

[9] Mert Yüksekgönül, Federico Bianchi, Joseph Boen, Sheng Liu, Pan Lu, Zhi Huang, Carlos Guestrin, and James Zou. Optimizing generative AI by backpropagating language model feedback. *Nat.*, 639(8055):609–616, 2025. doi: 10.1038/S41586-025-08661-4. URL https://doi.org/10.1038/S41586-025-08661-4.

[10] Peng Xia, Rujun Han, Zifeng Wang, Yanfei Chen, Yufan Zhang, Yoonho Lee, Chengsong Huang, Han Yu, Zhongying CuiZhu, Yifei Ming, Huaxiu Yao, Burak Gokturk, Tomas Pfister, and Chen-Yu Lee. RRSI: Regularized recursive self-improvement of agent harnesses. arXiv preprint arXiv:2609.24972, 2026. URL https://arxiv.org/abs/2609.24972.

[11] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-Refine: Iterative Refinement with Self-Feedback. arXiv:2303.17651, 2023. URL https://arxiv.org/abs/2303.17651.

[12] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R. Narasimhan, and Yuan Cao. ReAct: Synergizing Reasoning and Acting in Language Models. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. URL https://openreview.net/forum?id=WE_vluYUL-X.

[13] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language Agents with Verbal Reinforcement Learning. arXiv:2303.11366, 2023. URL https://arxiv.org/abs/2303.11366.

[14] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An Open-Ended Embodied Agent with Large Language Models. arXiv:2305.16291, 2023. URL https://arxiv.org/abs/2305.16291.

[15] Qizheng Zhang, Changran Hu, Shubhangi Upasani, Boyuan Ma, Fenglu Hong, Vamsidhar Kamanuru, Jay Rainton, Chen Wu, Mengmeng Ji, Hanchen Li, Urmish Thakker, James Zou, and Kunle Olukotun. Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models. arXiv:2510.04618, 2025. URL https://arxiv.org/abs/2510.04618.

[16] Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. Promptbreeder: Self-Referential Self-Improvement Via Prompt Evolution. arXiv:2309.16797, 2023. URL https://arxiv.org/abs/2309.16797.

[17] Jenny Zhang, Bingchen Zhao, Wannan Yang, Jakob Foerster, Jeff Clune, Minqi Jiang, Sam Devlin, and Tatiana Shavrina. Hyperagents. arXiv:2603.19461, 2026. URL https://arxiv.org/abs/2603.19461.

[18] Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning. arXiv:2507.19457, 2025. URL https://arxiv.org/abs/2507.19457.

[19] Minki Kang, Wei-Ning Chen, Dongge Han, Huseyin A. Inan, Lukas Wutschitz, Yanzhi Chen, Robert Sim, and Saravan Rajmohan. ACON: Optimizing Context Compression for Long-horizon LLM Agents. *CoRR*, abs/2510.00615, 2025. doi: 10.48550/ARXIV.2510.00615. URL https://doi.org/10.48550/ARXIV.2510.00615.

[20] Siru Ouyang, Jun Yan, I-Hung Hsu, Yanfei Chen, Ke Jiang, Zifeng Wang, Rujun Han, Long T. Le, Samira Daruki, Xiangru Tang, Vishy Tirumalashetty, George Lee, Mahsan Rofouei, Hangfei Lin, Jiawei Han, Chen-Yu Lee, and Tomas Pfister. ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory. arXiv:2509.25140, 2025. URL https://arxiv.org/abs/2509.25140.

[21] Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-MEM: Agentic Memory for LLM Agents. arXiv:2502.12110, 2025. URL https://arxiv.org/abs/2502.12110.

[22] Ruimin Pei, Yongkang Wu, Shangyi Zheng, Yaqing Zhang, Deyang Li, Jianjun Tao, Xinyu Zhang, and Xiang Zhang. TTSE: A Two-Track Online Self-Evolution Framework. arXiv preprint arXiv:2609.24289, 2026. URL https://arxiv.org/abs/2609.24289.

[23] Chongyu Qu, Can Cui, Zhengyi Lu, Junchao Zhu, Tianyuan Yao, Junlin Guo, Juming Xiong, Yanfan Zhu, Yuechen Yang, Bennett A. Landman, and Yuankai Huo. SAGEAgent: A Self-Evolving Agent for Cost-Aware Modality Acquisition in Multimodal Survival Prediction. arXiv preprint arXiv:2607.09521, 2026. URL https://arxiv.org/abs/2607.09521.

[24] Xuefei Julie Wang, Lauren Hyoseo Yoon, Chengrui Qu, Amanda Zichang Wang, Atharva Sehgal, Eric Mazumdar, and Yisong Yue. Knowledge-Centric Self-Improvement. arXiv preprint arXiv:2607.19592, 2026. URL https://arxiv.org/abs/2607.19592.

[25] Wenhao Wu, Menghao Zhang, Xin Wang, Zhi Wang, Kun Shao, and Jian Luan. TRACE: A Self-Evolving Skill Bank for Consistent, Limit-Aware LLM Agents. arXiv:2608.22793, 2026. URL https://arxiv.org/abs/2608.22793.

[26] Tianxin Wei, Zhan Shi, Minhua Lin, Bing He, Zewen Liu, Yisi Sang, Yuanchen Bei, Xuying Ning, Jiaru Zou, Ting-Wei Li, Xiao Lin, Yanjun Zhao, Chi Wang, Benoit Dumoulin, Dakuo Wang, Jingrui He, and Hanqing Lu. Evo-Harness: Context-to-Harness Skill Compilation for Self-Evolving Agents. arXiv:2608.15071, 2026. URL https://arxiv.org/abs/2608.15071.

[27] Yichun Feng, Jiawei Wang, and Haozhe Sun. A Wrong Turn Does Not Ruin the Journey: Deviation-Guided Skill Self-Evolution for LLM Agents. arXiv preprint arXiv:2609.29154, 2026. URL https://arxiv.org/abs/2609.29154.

[28] Yangbo Wei, Zhen Huang, Shaoqiang Lu, Junhong Qian, Qifan Wang, Chen Wu, and Lei He. SkillSmith: Co-Evolving Skills and Tools for Self-Improving Agent Systems. *CoRR*, abs/2606.01314, 2026. doi: 10.48550/ARXIV.2606.01314. URL https://doi.org/10.48550/ARXIV.2606.01314.

[29] Junhao Qiu, Qinglong Hu, Xialiang Tong, Mingxuan Yuan, Liyong Lin, and Qingfu Zhang. AlgoEvo: Self-Evolving Agentic Search for Automated Algorithm Discovery. arXiv preprint arXiv:2609.15820, 2026. URL https://arxiv.org/abs/2609.15820.

[30] Zhaochen Yu, Yingcheng Wu, Zhenfei Yin, Kaiyuan Chen, Zhe Zhao, Mengdi Wang, Shuicheng Yan, and Ling Yang. Recursive Experiential-Working Memory Evolution for Long-Horizon Agent Harnesses. arXiv:2608.24876, 2026. URL https://arxiv.org/abs/2608.24876.

[31] Wenyi Wang, Piotr Piekos, Nanbo Li, Firas Laakom, Yimeng Chen, Mateusz Ostaszewski, Mingchen Zhuge, and Jürgen Schmidhuber. Huxley-Gödel Machine: Human-Level Coding Agent Development by an Approximation of the Optimal Self-Improving Machine. *CoRR*, abs/2510.21614, 2025. doi: 10.48550/ARXIV.2510.21614. URL https://doi.org/10.48550/ARXIV.2510.21614.

[32] Changzhi Liu, Yilun Liu, Sikuan Yan, Volker Tresp, and Yunpu Ma. Mendel Gödel Machine: Recursive Self-Improving Coding Agents via Comparative Evolution. *CoRR*, abs/2608.07645, 2026. doi: 10.48550/ARXIV.2608.07645. URL https://doi.org/10.48550/ARXIV.2608.07645.

[33] Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. Agent Workflow Memory. arXiv:2409.07429, 2024. URL https://arxiv.org/abs/2409.07429.

[34] Yang Xiao, Yusong Sun, Haoyi Wu, Wenyang Hui, Wen Da, Zhaokai Luo, Mu Chuan, Yao Hu, Wenjie Li, and Chengyue Jiang. PILOT in the Loop: Live Self-Improvement for Long-Horizon Agents. arXiv preprint arXiv:2608.26530, 2026. URL https://arxiv.org/abs/2608.26530.

[35] Xinghong Fu, Aravinth Kulanthaivelu, and Yutaro Yamada. Self improvement via fast tree-search. arXiv preprint arXiv:2609.19526, 2026. URL https://arxiv.org/abs/2609.19526.

[36] Yuru Feng, Yaoqi Chen, Beidi Zhao, Qianxi Zhang, Xinjiang Wang, Jianan Lu, Zhirui Wang, Shusen Xu, Zengzhong Li, and Qi Chen. SkillHEX: Improving Agent Skills via Hypothesis-Driven Autonomous Exploration and Exploitation. arXiv preprint arXiv:2608.05628, 2026. URL https://arxiv.org/abs/2608.05628.

[37] Hongyang Du, Lan Yan, Christian Flores, and Asim Kadav. Designer-RSI: Evolving Procedural Memory from User Traffic for Agentic Graphic Design. arXiv preprint arXiv:2609.22086, 2026. URL https://arxiv.org/abs/2609.22086.

[38] Mengzhuo Chen, Junjie Wang, Zhe Liu, Yawen Wang, Haiming Zheng, and Qing Wang. From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. arXiv:2606.06324, 2026. URL https://arxiv.org/abs/2606.06324.

[39] Guhong Chen, Yingcheng Shi, Yongbin Li, Binhua Li, Xander Xu, Hu Wei, Shiwen Ni, Min Yang, and Jieping Ye. EvoTrainer: Co-Evolving LLM Policies and Training Harnesses for Autonomous Agentic Reinforcement Learning. *CoRR*, abs/2606.03108, 2026. doi: 10.48550/arXiv.2606.03108. URL https://arxiv.org/abs/2606.03108.

[40] Ruiqing Yue, Yu Cui, Zhuoyu Sun, Sicheng Pan, Xianhong Xue, Tingyu Li, Ting Li, Wenzhuo Zhu, Yi Chen, Yifei Liu, Baohan Huang, Zhe Cui, Haibin Zhang, and Cong Zuo. Ecdysis: Efficient and Effective Training of Runtime Harnesses for LLM Agents. arXiv preprint arXiv:2609.11677, 2026. URL https://arxiv.org/abs/2609.11677.

[41] Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. In *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, 2015. URL http://arxiv.org/abs/1412.6980.

[42] David E. Rumelhart, Geoffrey E. Hinton, and Ronald J. Williams. Learning representations by back-propagating errors. *Nature*, 323(6088):533–536, 1986. doi: 10.1038/323533a0. URL https://doi.org/10.1038/323533a0.

[43] Richard S. Sutton and Andrew G. Barto. *Reinforcement learning - an introduction, 2nd Edition.* MIT Press, 2018. URL http://www.incompleteideas.net/book/the-book-2nd.html.

[44] Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks. In *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, volume 70 of *Proceedings of Machine Learning Research*, pages 1126–1135. PMLR, 2017. URL http://proceedings.mlr.press/v70/finn17a.html.

[45] Marcin Andrychowicz, Misha Denil, Sergio Gomez Colmenarejo, Matthew W. Hoffman, David Pfau, Tom Schaul, and Nando de Freitas. Learning to learn by gradient descent by gradient descent. In *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*, pages 3981–3989, 2016. URL https://proceedings.neurips.cc/paper/2016/hash/fb87582825f9d28a8d42c5e5e5e8b23d-Abstract.html.

[46] Geoffrey E. Hinton, Oriol Vinyals, and Jeffrey Dean. Distilling the Knowledge in a Neural Network. *CoRR*, abs/1503.02531, 2015. URL http://arxiv.org/abs/1503.02531.

[47] Yang Li, Semih Yavuz, and Shafiq Joty. RISE: Recursive improvement via self-extrapolating policy distillation. arXiv preprint arXiv:2609.05295, 2026. URL https://arxiv.org/abs/2609.05295.

[48] Christos Louizos, Max Welling, and Diederik P. Kingma. Learning Sparse Neural Networks through L_0 Regularization. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, 2018. URL https://openreview.net/forum?id=H1Y8hhg0b.

[49] Ian J. Goodfellow, Yoshua Bengio, and Aaron C. Courville. *Deep Learning*. Adaptive computation and machine learning. MIT Press, 2016. ISBN 978-0-262-03561-3. URL https://www.deeplearningbook.org/.

[50] Nitish Srivastava, Geoffrey E. Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1):1929–1958, 2014. URL https://jmlr.org/papers/v15/srivastava14a.html.

[51] Guanyu Nie, Fangzhou Zhu, Shixiong Kai, Xiongwei Han, Tao Zhong, and Mingxuan Yuan. SkillEvoReg: Regularizing Agent Skill Evolution Against Overfitting. arXiv preprint arXiv:2609.30861, 2026. URL https://arxiv.org/abs/2609.30861.

[52] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. ImageNet Classification with Deep Convolutional Neural Networks. In *Advances in Neural Information Processing Systems 25*, pages 1106–1114, 2012. URL https://papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks.

[53] Chengyi Yang, Zhishang Xiang, Yunbo Tang, Zongpei Teng, Chengsong Huang, Fei Long, Yuhan Liu, and Jinsong Su. TTCS: Test-Time Curriculum Synthesis for Self-Evolving. arXiv preprint arXiv:2601.22628, 2026. URL https://arxiv.org/abs/2601.22628.

[54] Sylvestre-Alvise Rebuffi, Alexander Kolesnikov, Georg Sperl, and Christoph H. Lampert. iCaRL: Incremental classifier and representation learning. In *2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017*, pages 5533–5542. IEEE Computer Society, 2017. doi: 10.1109/CVPR.2017.587. URL https://doi.org/10.1109/CVPR.2017.587.

[55] Zhongxin Huang, Songyang Li, Renzhe Zhou, Feiran Zhu, Chenglei Dai, Zhen Xiao, Xuanping Li, and Jingwei Zhuo. SEEK: Skill-Routed Evaluation with Evolvable Knowledge for Industrial Search. arXiv preprint arXiv:2609.29803, 2026. URL https://arxiv.org/abs/2609.29803.

[56] Song Han, Huizi Mao, and William J. Dally. Deep Compression: Compressing Deep Neural Networks with Pruning, Trained Quantization and Huffman Coding. arXiv:1510.00149, 2015. URL https://arxiv.org/abs/1510.00149.

[57] Mingxuan Zheng, Yujin Zhou, Chuxue Cao, Boqin Yin, Yuyao Zhang, Jiapeng Sun, Shuaishuai Gong, Sirui Han, and Yike Guo. SkillProx: Self-Evolving Agent Skills via Proximal Textual Gradient Descent. arXiv preprint arXiv:2608.07449, 2026. URL https://arxiv.org/abs/2608.07449.

[58] Thomas G. Dietterich. Ensemble methods in machine learning. In *Multiple Classifier Systems*, volume 1857 of *Lecture Notes in Computer Science*, pages 1–15. Springer, 2000. doi: 10.1007/3-540-45014-9_1. URL https://doi.org/10.1007/3-540-45014-9_1.

[59] Xingjian Wu, Junkai Lu, Siyu Yan, Xiangfei Qiu, Jilin Hu, Chenjuan Guo, and Bin Yang. Differentiable Mixture-of-Agents Incentivizes Swarm Intelligence of Large Language Models. arXiv preprint arXiv:2605.15706, 2026. URL https://arxiv.org/abs/2605.15706.

[60] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron C. Courville, and Yoshua Bengio. Generative Adversarial Nets. In *Advances in Neural Information Processing Systems 27: Annual Conference on Neural Information Processing Systems 2014, December 8-13 2014, Montreal, Quebec, Canada*, pages 2672–2680, 2014. URL https://proceedings.neurips.cc/paper/2014/hash/f033ed80deb0234979a61f95710dbe25-Abstract.html.

[61] Yixu Huang, Xinglei Yu, and Zhongyu Wei. ACE: Self-Evolving LLM Coding Framework via Adversarial Unit Test Generation and Preference Optimization. *CoRR*, abs/2605.16299, 2026. doi: 10.48550/arXiv.2605.16299. URL https://arxiv.org/abs/2605.16299.

[62] Jonathan Frankle and Michael Carbin. The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net, 2019. URL https://openreview.net/forum?id=rJl-b3RcF7.

[63] Sungrae Park, Jun-Keon Park, Su-Jin Shin, and Il-Chul Moon. Adversarial Dropout for Supervised and Semi-Supervised Learning. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, New Orleans, Louisiana, USA, February 2-7, 2018*, pages 3917–3924. AAAI Press, 2018. doi: 10.1609/aaai.v32i1.11634. URL https://ojs.aaai.org/index.php/AAAI/article/view/11634.

[64] Shibhansh Dohare, J. Fernando Hernandez-Garcia, Qingfeng Lan, Parash Rahman, A. Rupam Mahmood, and Richard S. Sutton. Loss of plasticity in deep continual learning. *Nat.*, 632(8026):768–774, 2024. doi: 10.1038/s41586-024-07711-7. URL https://doi.org/10.1038/s41586-024-07711-7.

[65] Kevin Swersky, Jasper Snoek, and Ryan Prescott Adams. Freeze-thaw bayesian optimization. *CoRR*, abs/1406.3896, 2014. URL https://arxiv.org/abs/1406.3896.

[66] Siwei Wu, Jincheng Ren, Yizhi Li, Haau-Sing Li, Chengran Yang, Yuxuan Zhang, Weicheng Gu, Jian Yang, Riza Batista-Navarro, Chuanyi Zhang, Xianglong Liu, Ming Zhou, Bryan Dai, and Chenghua Lin. ModularRSI: Modular and Generalizable Recursive Harness Self-Improvement. arXiv preprint arXiv:2609.14857, 2026. URL https://arxiv.org/abs/2609.14857.

[67] Nadav Kashtan and Uri Alon. Spontaneous evolution of modularity and network motifs. *Proceedings of the National Academy of Sciences*, 102(39):13773–13778, 2005. doi: 10.1073/pnas.0503610102. URL https://doi.org/10.1073/pnas.0503610102.

[68] Tianxin Wei, Noveen Sachdeva, Benjamin Coleman, Zhankui He, Yuanchen Bei, Xuying Ning, Mengting Ai, Yunzhe Li, Jingrui He, Ed H. Chi, Chi Wang, Shuo Chen, Fernando Pereira, Wang-Cheng Kang, and Derek Zhiyuan Cheng. Evo-Memory: Benchmarking LLM Agent Test-time Learning with Self-Evolving Memory. *CoRR*, abs/2511.20857, 2025. doi: 10.48550/ARXIV.2511.20857. URL https://doi.org/10.48550/ARXIV.2511.20857.

[69] Ziao Zhang, Kou Shi, Shiting Huang, Avery Nie, Yu Zeng, Yiming Zhao, Zhen Fang, Qishen Su, Haibo Qiu, Wei Yang, Qingnan Ren, Shun Zou, Wenxuan Huang, Lin Chen, Zehui Chen, and Feng Zhao. SkillFlow: Benchmarking Lifelong Skill Discovery and Evolution for Autonomous Agents. arXiv:2604.17308, 2026. URL https://arxiv.org/abs/2604.17308.

[70] Dong Yan, Jian Liang, Dapeng Hu, Ran He, Nicholas Jing Yuan, Qi Zhang, and Tieniu Tan. AgentStream: How Well Do Self-Evolving LLM Agents Perform Under Streaming Tasks? arXiv:2608.00155, 2026. URL https://arxiv.org/abs/2608.00155.

[71] Xidong Yang, Xingyi Zhang, Wenhao Li, Wenyan Liu, Junjie Sheng, Yun Hua, Wei Yin, Tao Fang, Chuyun Shen, and Xiangfeng Wang. PATH-Bench: Path-Dependent Evaluation of Lifelong Agents. arXiv:2608.01149, 2026. URL https://arxiv.org/abs/2608.01149.

[72] Qingyao Ai, Yichen Tang, Changyue Wang, Jianming Long, Weihang Su, and Yiqun Liu. MemoryBench: A Benchmark for Memory and Continual Learning in LLM Systems. *CoRR*, abs/2510.17281, 2025. doi: 10.48550/ARXIV.2510.17281. URL https://doi.org/10.48550/ARXIV.2510.17281.

[73] Leijun Zhou, Zhihao Liu, Xiang Qu, Chenxu Liu, Yifei Liu, Yanke Yu, Jingzhe Xu, Xuejun Wu, Buyue Qian, Xi Chen, Yaowei Zheng, and Junhao Hu. GDPevo: Evaluating Agent Self-Evolution on Real Business Tasks. arXiv:2608.03764, 2026. URL https://arxiv.org/abs/2608.03764.

[74] Varun Ursekar, Apaar Shanker, Yash Maurya, Shehab Yasser, Vijay S. Kalmath, Veronica Chatrath, and Yuan Xue. HarnessOpt-Bench: Evaluating LLMs at Harness Optimization. arXiv:2608.06301, 2026. URL https://arxiv.org/abs/2608.06301.

[75] Jiajun Shi, Siyuan Tao, Yuhao Wu, Zexuan Wang, Jingyuan Zhang, Jiaheng Liu, Xinping Lei, Xinrong Zhang, Siyuan Fang, Zhewen Tan, Tianle Cai, Junhao Fang, Jiameng Huang, Yueyang Wang, Jinkai Liu, Yuxuan Zhang, Jian Yang, Zhoujun Li, Shen Yan, Wenhao Huang, and Ge Zhang. S3Gym: Can LLMs Turn Self-Testing and Self-Judging into Self-Improvement? arXiv:2608.31100, 2026. URL https://arxiv.org/abs/2608.31100.

[76] Zihao Cheng, Zeming Liu, Yingyu Shan, Xinyi Wang, Xiangrong Zhu, Yunpu Ma, Hongru Wang, Yuhang Guo, Wei Lin, and Yunhong Wang. Mem^{2}Evolve: Towards Self-Evolving Agents via Co-Evolutionary Capability Expansion and Experience Distillation. arXiv:2604.10923, 2026. URL https://arxiv.org/abs/2604.10923.

[77] Tianyi Guan, Yiding Wang, Haotong Yang, Siyuan Cao, Shirui Liu, Yi Hu, Jiaqi Li, and Muhan Zhang. ContinualSkillBench: Can LLM Agents Truly Evolve Their Capabilities? arXiv:2608.03874, 2026. URL https://arxiv.org/abs/2608.03874.

[78] Fanqing Meng, Lingxiao Du, Qiguang Chen, Ziqi Zhao, Haocheng Lu, Mengkang Hu, and Michael Qizhe Shieh. RSIBench-Data: Benchmarking Data-Centric Research for Recursive Self-Improvement. arXiv:2607.25886, 2026. URL https://arxiv.org/abs/2607.25886.

[79] Xingze Gao, Chuanrui Hu, Hongda Chen, Pengfei Yao, Zhao Wang, Yi Bai, Zhengwei Wu, Yunyun Han, Xiaofeng

Cong, Jie Gui, Yafeng Deng, and Teng Li. EvoAgentBench: Benchmarking Agent Self-Evolution via Ability Transfer. arXiv:2607.05202, 2026. URL https://arxiv.org/abs/2607.05202.

[80] Yuhao Wu, Jingyuan Zhang, Jiajun Shi, Xinping Lei, Qingshui Gu, Yuxuan Zhang, Zexuan Wang, Chen He, Chen Huang, Maojia Song, Zhiyuan Zeng, Shaowen Wang, Jinkai Liu, Yunfeng Shi, Jiaheng Liu, Shen Yan, Wenhao Huang, Ge Zhang, and Wenxuan Zhang. HarnessDev: Can LLMs Create and Evolve Their Own Agent Harness? arXiv:2609.01437, 2026. URL https://arxiv.org/abs/2609.01437.

[81] Lisheng Huang, Chen Yang, Hao Zhou, Huatong Song, Zongchao Chen, Ran Le, Yang Song, Wayne Xin Zhao, and Tao Zhang. Evo-Bench: Can Language Models Improve Agent Harness? arXiv:2608.09096, 2026. URL https://arxiv.org/abs/2608.09096.

[82] Jiahang Lin, Shichun Liu, Chengjun Pan, Lizhi Lin, Shihan Dou, Zhiheng Xi, Xuanjing Huang, Hang Yan, Zhenhua Han, Tao Gui, and Yu-Gang Jiang. Agentic Harness Engineering: Observability-Driven Automatic Evolution of Coding-Agent Harnesses. arXiv:2604.25850, 2026. URL https://arxiv.org/abs/2604.25850.

[83] Dhruv Srikanth, Bingchen Zhao, Dixing Xu, Yuxiang Wu, and Zhengyao Jiang. Recursive self-improvement of AI research agents. arXiv preprint arXiv:2609.26457, 2026. URL https://arxiv.org/abs/2609.26457.

[84] Tong Zheng, Haolin Liu, Chengsong Huang, Huiwen Bao, Sheng Zhang, Rui Liu, Runpeng Dai, Ruibo Chen, Chenxi Liu, Tianyi Xiong, Xidong Wu, Hongming Zhang, and Heng Huang. LLMs Improving LLMs: Agentic Discovery for Test-Time Scaling. arXiv preprint arXiv:2605.08083, 2026. URL https://arxiv.org/abs/2605.08083.

[85] Yongshi Ye, Tian Lan, Feihu Jiang, Muyang Ye, Bin Zhu, Qianghuai Jia, Longyue Wang, Zhao Xu, Weihua Luo, and Xiaodong Shi. CHIME: Credit-Aware Hierarchical Memory Evolution for Long-Horizon Agentic Planning. arXiv preprint arXiv:2609.02074, 2026. URL https://arxiv.org/abs/2609.02074.

[86] Pingchen Lu, Xiangyi Wang, Xiang Li, Jie Mao, Zikun Qu, Junfeng Luo, Yao Shu, Bryan Kian Hsiang Low, and Zhongxiang Dai. COBRA-Skills: Contextual Bandit-Guided Evolution for Agent Skill Optimization. arXiv preprint arXiv:2609.11682, 2026. URL https://arxiv.org/abs/2609.11682.

[87] Seth Karten, Joel Zhang, Tersoo Upaa Jr, Ruirong Feng, Wenzhe Li, Chengshuai Shi, Chi Jin, and Kiran Vodrahalli. Continual Harness: Online Adaptation for Self-Improving Foundation Agents. arXiv:2605.09998, 2026. URL https://arxiv.org/abs/2605.09998.

[88] Weihao Tan, Wentao Zhang, Xinrun Xu, Haochong Xia, Ziluo Ding, Boyu Li, Bohan Zhou, Junpeng Yue, Jiechuan Jiang, Yewen Li, Ruyi An, Molei Qin, Chuqiao Zong, Longtao Zheng, Yujie Wu, Xiaoqiang Chai, Yifei Bi, Tianbao Xie, Pengjie Gu, Xiyun Li, Ceyao Zhang, Long Tian, Chaojie Wang, Xinrun Wang, Börje F. Karlsson, Bo An, Shuicheng Yan, and Zongqing Lu. Cradle: Empowering Foundation Agents Towards General Computer Control. arXiv:2403.03186, 2024. URL https://arxiv.org/abs/2403.03186.

[89] Yifan Zhang, Yutong Dai, Juntao Tan, Luyu Yang, Rishi Mullur, Thai Hoang, Zhiyuan Hu, James Zhu, Phil Mui, Silvio Savarese, Ran Xu, and Zeyuan Chen. DarwinX: Evolving Agent Harnesses Through Natural Selection. arXiv:2608.07545, 2026. URL https://arxiv.org/abs/2608.07545.

[90] Shangjian Yin, Zehao Zhao, Kavosh Asadi, Rui Liu, Yuchen Lu, Shike Mei, Hang Cui, Luke Simon, Zhouxing Shi, and Hamed Firooz. Recursive Self-Improvement via On-Policy Distillation for Reasoning. arXiv preprint arXiv:2609.30652, 2026. URL https://arxiv.org/abs/2609.30652.

[91] Lirong Che, Yuzhe yang, Peiwen lin, Chuang wang, Xueqian wang, and Jian su. DemoEvolve: Overcoming Sparse Feedback in Agentic Harness Evolution with Demonstrations. arXiv:2605.24539, 2026. URL https://arxiv.org/abs/2605.24539.

[92] Siheng Xiong, Ali Payani, Oguzhan Gungordu, and Faramarz Fekri. DIVE: Unlocking Self-Improvement in Frozen Language Models Through Diversity-Driven Skill Evolution. arXiv:2608.12486, 2026. URL https://arxiv.org/abs/2608.12486.

[93] Tong Zheng, Xidong Wu, Zheng Zhang, Zhankui He, Chaoyi Zhang, Benjamin Coleman, Ruoqiao Wei, Di Bai, Haolin Liu, Rui Liu, Xue Wang, Yue Zhuan, Wang-Cheng Kang, Renkai Xiang, Heng Huang, Xinwu Cheng, and Yunsong Guo. Dream-RSI: Recursive Self-Improvement through Evolving Worlds, 2026. URL https://dream-rsi.com/assets/dream-rsi.pdf. Project manuscript.

[94] Mirac Suzgun, Mert Yuksekgonul, Federico Bianchi, Dan Jurafsky, and James Zou. Dynamic Cheatsheet: Test-Time Learning with Adaptive Memory. arXiv:2504.07952, 2025. URL https://arxiv.org/abs/2504.07952.

[95] Chengsong Huang, Zifeng Wang, Rujun Han, Jun Yan, Yanfei Chen, Zoey CuiZhu, Ke Jiang, Peng Xia, Han Yu, Yufan Zhuang, Yifei Ming, Jiaqi Pan, Bhavana Dalvi Mishra, Jiaxin Huang, Burak Gokturk, Tomas Pfister, and Chen-Yu Lee. EnvHarness: Awakening Static Worlds for Agent Learning. arXiv preprint arXiv:2608.19880, 2026. URL https://arxiv.org/abs/2608.19880.

[96] Ziyang Liu, Xinyan Guo, Xuchen Wei, Han Hao, and Liu Yang. Escher-Loop: Mutual Evolution by Closed-Loop Self-Referential Optimization. arXiv:2604.23472, 2026. URL https://arxiv.org/abs/2604.23472.

[97] Jun Nie, Yonggang Zhang, Qianshu Cai, Yiu-ming Cheung, Xinmei Tian, and Bo Han. EvolveNet: Collaborative Harness Evolution for Agent Self-Improvement. arXiv:2608.04968, 2026. URL https://arxiv.org/abs/2608.04968.

[98] Sehee Kim, Yumin Choi, Minki Kang, and Sung Ju Hwang. EvolveTrade: Experience-Driven Policy Refinement for Self-Evolving LLM Trading Agents. arXiv preprint arXiv:2609.17632, 2026. URL https://arxiv.org/abs/2609.17632.

[99] Meiduo Chong, Shaolei Zhang, Ju Fan, and Xiaoyong Du. EvoOntology: A Self-Evolving Ontology Layer for Data Agents. arXiv preprint arXiv:2609.15779, 2026. URL https://arxiv.org/abs/2609.15779.

[100] Bofan Chen, Boxuan Zhang, Fei Tang, Zhengxi Lu, Yong Du, Tongbo Chen, Weiming Lu, Jun Xiao, Yueting Zhuang, and Yongliang Shen. Reflect, Revise, Reuse: Training-Free Skill Evolution for GUI Agents. arXiv preprint arXiv:2609.17653, 2026. URL https://arxiv.org/abs/2609.17653.

[101] Wenbo Gao, Zhaomou Song, Zhiyuan Ji, Renxi Liu, Xing Li, Xianzhi Yu, Xiaoguang Li, James Chung-wai Cheung, Weizhe Lin, and Yaoyuan Wang. Experience Funnel: A State-Policy Alternating Loop for Self-Evolving Agents. arXiv preprint arXiv:2609.08919, 2026. URL https://arxiv.org/abs/2609.08919.

[102] Tao Feng, Chongrui Ye, Tianyang Luo, Jingjun Xu, Xueqiang Xu, Haozhen Zhang, Zhigang Hua, Yan Xie, Shuang Yang, Ge Liu, and Jiaxuan You. ExpGraph: Model-Agnostic Experience Learning with Graph-Structured Memory for LLM Agents. arXiv:2605.30712, 2026. URL https://arxiv.org/abs/2605.30712.

[103] Linhao Zhong, Zongze Du, Linyu Wu, Yu Bo, Hourong Li, Chenchen Jing, Hao Chen, Yuling Xi, and Chunhua Shen. ForeDreamer: A Self-Evolving Dual-Agent Memory Architecture for Future Event Prediction. arXiv preprint arXiv:2608.20920, 2026. URL https://arxiv.org/abs/2608.20920.

[104] Chengsong Huang, Haolin Liu, Tong Zheng, Runpeng Dai, Langlin Huang, Jinyuan Li, Zongxia Li, Zhepei Wei, Yu Meng, and Jiaxin Huang. G-Zero: Self-Play for Open-Ended Generation from Zero Data. arXiv preprint arXiv:2605.09959, 2026. URL https://arxiv.org/abs/2605.09959.

[105] Zhaotian Weng, Antonis Antoniades, Deepak Nathani, Zhen Zhang, Xiao Pu, and Xin Eric Wang. Group-Evolving Agents: Open-Ended Self-Improvement via Experience Sharing. arXiv:2602.04837, 2026. URL https://arxiv.org/abs/2602.04837.

[106] Laizhen Li, Jiarui Li, Juanjuan Zhao, Kejiang Ye, Ye Li, Cheng-zhong Xu, and Xitong Gao. Grow the Harness, Not the Context: From Strategy-Free Scaffolds to Reusable Specialist Agents. arXiv preprint arXiv:2609.26760, 2026. URL https://arxiv.org/abs/2609.26760.

[107] Xunjian Yin, Xinyi Wang, Liangming Pan, Li Lin, Xiaojun Wan, and William Yang Wang. Gödel Agent: A Self-Referential Agent Framework for Recursive Self-Improvement. arXiv:2410.04444, 2024. URL https://arxiv.org/abs/2410.04444.

[108] Haoyang Yan, Min-le Su, Hangfan Zhang, Zhanhao Li, Chen Zhang, Shao Zhang, Yang Chen, Lei Bai, and Shuyue Hu. Harness-of-Harness: Multi-Day Autonomous Software Development with Continual Improvement. arXiv preprint arXiv:2609.01481, 2026. URL https://arxiv.org/abs/2609.01481.

[109] Haoran Ye, Yuxing Lu, Haonan Dong, Zhaochen Su, and Guojie Song. Harness-Zero: Harness Distillation via Agent-as-Harness. arXiv preprint arXiv:2609.24974, 2026. URL https://arxiv.org/abs/2609.24974.

[110] Xiaotian Luo, Dizhan Xue, Fengxingyu Wang, Chuanrui Hu, and Yafeng Deng. HarnessBank: Semantic Gene-Bank Search with Gated Verification for Agent-Harness Self-Evolution. arXiv:2607.13683, 2026. URL https://arxiv.org/abs/2607.13683.

[111] Wen Jiang, Mingmin Chu, Yimeng Tian, Qianxin Zhang, Haofei Yang, Rui Yang, Yang Liu, Tao Lv, and Fangming Li. HarnessEvolve: Learning from Reference Trajectories for Reliable Agent Self-Evolution. arXiv:2609.00829, 2026. URL https://arxiv.org/abs/2609.00829.

[112] Xin Wang, Wenhao Wu, Menghao Zhang, Zhi Wang, Kun Shao, Jian Luan, Yang Li, Qing Li, Shangding Gu, Huichi Zhou, Shuqing Shi, Fei Ni, Shuo Lu, Weicheng Meng, Kang Li, Jin Wu, Kang Zhao, Shangmin Guo, Gen Li, Yongqiang Tang, Zhizhong Zhang, Yuan Xie, and Heng Qu. HarnessPAI: An Evolving Harness for Physical AI. arXiv preprint arXiv:2609.29166, 2026. URL https://arxiv.org/abs/2609.29166.

[113] Tingyang Chen, Shuo Lu, Kang Zhao, Weicheng Meng, Hanlin Teng, Tianhao Li, Chao Li, Xule Liu, Jian Liang, Zhizhong Zhang, Yuan Xie, Heng Qu, Kun Shao, and Jian Luan. HarnessX: A Composable, Adaptive, and Evolvable Agent Harness Foundry. arXiv:2606.14249, 2026. URL https://arxiv.org/abs/2606.14249.

[114] Jinchang Zhu, Jindong Li, Cheng Zhang, Jiahong Liu, and Menglin Yang. HeLa-Mem: Hebbian Learning and Associative Memory for LLM Agents. arXiv:2604.16839, 2026. URL https://arxiv.org/abs/2604.16839.

[115] Ruiyao Xu, Tiankai Yang, and Wei-Chieh Huang. HyperSkill: Self-Evolving LLM Agents via Hypergraph-Structured Skill Memory. arXiv:2608.16114, 2026. URL https://arxiv.org/abs/2608.16114.

[116] Prakhar Dixit and Tim Oates. ISM:Self-Improving Strategy Memory for Continual Mathematical Reasoning. arXiv:2606.31191, 2026. URL https://arxiv.org/abs/2606.31191.

[117] Zhening Li, Joshua Liu, Mateja Vukelic, Nicole Shen, Supriya Lall, Amitayush Thakur, Alex Zhang, Omar Khattab, Jonathan Light, and Armando Solar-Lezama. Harness as a Language: A Minimalist Agent Framework With Maximal Expressivity. arXiv preprint arXiv:2609.26891, 2026. URL https://arxiv.org/abs/2609.26891.

[118] Haoran Ye, Xuning He, Vincent Arak, Haonan Dong, and Guojie Song. Meta Context Engineering via Agentic Skill Evolution. arXiv preprint arXiv:2601.21557, 2026. URL https://arxiv.org/abs/2601.21557.

[119] Ruoyu Wu, Shenfu Xie, Yinqian Sun, Haibo Tong, and Feifei Zhao. MediSkill-Evo: Process-Constrained Self-Evolution for Evidence-Grounded Clinical Interaction. arXiv:2608.23397, 2026. URL https://arxiv.org/abs/2608.23397.

[120] Runnan Fang, Yuan Liang, Xiaobin Wang, Jialong Wu, Shuofei Qiao, Pengjun Xie, Fei Huang, Huajun Chen, and Ningyu Zhang. Memp: Exploring Agent Procedural Memory. arXiv:2508.06433, 2025. URL https://arxiv.org/abs/2508.06433.

[121] Yoonho Lee, Roshen Nair, Qizheng Zhang, Kangwook Lee, Omar Khattab, and Chelsea Finn. Meta-Harness: End-to-End Optimization of Model Harnesses. arXiv:2603.28052, 2026. URL https://arxiv.org/abs/2603.28052.

[122] Zongxia Li, Hongyang Du, Chengsong Huang, Xiyang Wu, Lantao Yu, Yicheng He, Jing Xie, Xiaomin Wu, Zhichao Liu, Jiarui Zhang, and Fuxiao Liu. MM-Zero: Self-Evolving Multi-Model Vision Language Models From Zero Data. arXiv preprint arXiv:2603.09206, 2026. URL https://arxiv.org/abs/2603.09206.

[123] Subhojyoti Mukherjee and Md Mehrab Tanjim. MoMHa: Multi-Objective Optimization of LLM Harnesses over Accuracy, Safety, and Tokens. arXiv preprint arXiv:2609.30967, 2026. URL https://arxiv.org/abs/2609.30967.

[124] Qianshu Cai, Yonggang Zhang, Xianzhang Jia, Huajiang Zheng, Wei Xue, Jun Song, Xinmei Tian, and Yike Guo. MOSS: Self-Evolution through Source-Level Rewriting in Autonomous Agent Systems. arXiv:2605.22794, 2026. URL https://arxiv.org/abs/2605.22794.

[125] Xunyi Zhao, Jian Zhou, Sihao Lin, Gengze Zhou, Zerui Li, Xinyu Yan, Jiajun Liu, Anton van den Hengel, and Qi Wu. NavHarness: Towards Lifelong Embodied Navigation. arXiv preprint arXiv:2609.34276, 2026. URL https://arxiv.org/abs/2609.34276.

[126] NeoHorse Team, Guoliang Cao, Guohao Dai, Tianyu Guo, Kai Han, Hailin Hu, Zihan Jiang, Xiang Kuang, Boxun Li, Yulong Li, Zehua Pei, Yuchuan Tian, Jiamin Wang, Yu Wang, Yunhe Wang, Yihong Wu, Haiyang Xu, Shuo Zhang, Hang Zhou, Siyang Cheng, Jiayu Fan, Wei He, Qingrui Jiao, Hongguang Li, Zhiyuan Li, Runke Liu, Xi Liu, Xinchen Liu, Sinno Jialin Pan, Yi Ren, Liuyang Song, Chenyu Wang, Bei Yu, Quanlu Zhang, Xiangyu Zhang, Mengyu Zheng, and Yingjie Zong. NeoHorse-1: Towards Recursive Self-Improvement via Agentic Post-Training with Routing Harness. arXiv preprint arXiv:2609.08183, 2026. URL https://arxiv.org/abs/2609.08183.

[127] Zhiling Yan, Dingjie Song, Hanrong Zhang, Wei Liang, Yuxuan Zhang, Yutong Dai, Lifang He, Philip S. Yu, Ran Xu, Xiang Li, and Lichao Sun. OpenSkill: Open-World Self-Evolution for LLM Agents. arXiv preprint arXiv:2606.06741, 2026. URL https://arxiv.org/abs/2606.06741.

[128] Xinyu Lu, Kaiqi Zhang, Jinglin Yang, Boxi Cao, Yaojie Lu, Hongyu Lin, Min He, Xianpei Han, and Le Sun. P^2O: Joint Policy and Prompt Optimization. arXiv:2603.21877, 2026. URL https://arxiv.org/abs/2603.21877.

[129] Yubo Li, Yidi Miao, Yuntian Shen, and Yuxin Liu. PANDO: Efficient Multimodal AI Agents via Online Skill Distillation. arXiv:2605.24785, 2026. URL https://arxiv.org/abs/2605.24785.

[130] Seth Karten, Alex L. Zhang, Kevin Thomas, Sebastian Müller, Elie Bakouch, Daniel Auras, Mika Senghaas, Fares Obeid, Konstantin Dunas, Johannes Hagemann, and Sami Jaghouar. Prime Agent: A Self-Improving RLM Harness. arXiv:2608.23552, 2026. URL https://arxiv.org/abs/2608.23552.

[131] Yuxing Lu, Yicheng Chen, Shanchan Wu, and Sercan Ö. Arık. Procedural Graphs: Self-Evolving Execution Structures for LLM Agents. arXiv preprint arXiv:2609.09153, 2026. URL https://arxiv.org/abs/2609.09153.

[132] Wenhao Yu, Zhenwen Liang, Chengsong Huang, Kishan Panaganti, Tianqing Fang, Haitao Mi, and Dong Yu. Guided Self-Evolving LLMs with Minimal Human Supervision. arXiv preprint arXiv:2512.02472, 2025. URL https://arxiv.org/abs/2512.02472.

[133] Chengsong Huang, Wenhao Yu, Xiaoyang Wang, Hongming Zhang, Zongxia Li, Ruosen Li, Jiaxin Huang, Haitao Mi, and Dong Yu. R-Zero: Self-Evolving Reasoning LLM from Zero Data. arXiv preprint arXiv:2508.05004, 2025. URL https://arxiv.org/abs/2508.05004.

[134] Alex Iacob, Andrej Jovanović, William F. Shen, Daniel Burkhardt, Meghdad Kurmanji, Nurbek Tastan, Lorenzo Sani, Niccolò Alberto Elia Venanzi, Ambroise Odonnat, Zeyu Cao, Bill Marino, Xinchi Qiu, and Nicholas D. Lane. The Red Queen Gödel Machine: Co-Evolving Agents and Their Evaluators. arXiv:2606.26294, 2026. URL https://arxiv.org/abs/2606.26294.

[135] Hyunin Lee, Jinglue Xu, Jeffrey Seely, Donghyun Lee, Matei Zaharia, and Yujin Tang. Recursive Harness Self-Improvement. arXiv:2607.15524, 2026. URL https://arxiv.org/abs/2607.15524.

[136] Sibo Zhu, Shicheng Fan, Xinyue Wang, Wenyi Wu, Kun Zhou, and Biwei Huang. RSIAgent: Autonomous Exploration for Recursive Self-improvement in New Environments. arXiv preprint arXiv:2609.15364, 2026. URL https://arxiv.org/abs/2609.15364.

[137] Hangfan Zhang, Shao Zhang, Kangcong Li, Chen Zhang, Yang Chen, Yiqun Zhang, Lei Bai, and Shuyue Hu. Self-Harness: Harnesses That Improve Themselves. arXiv:2606.09498, 2026. URL https://arxiv.org/abs/2606.09498.

[138] Wangcheng Tao, Han Wu, and Weng-Fai Wong. SePO: Self-Evolving Prompt Agent for System Prompt Optimization. arXiv:2606.04465, 2026. URL https://arxiv.org/abs/2606.04465.

[139] Qi Liu, Qinzheng Wang, Can Li, Yiming Bie, and Wanjing Ma. SimSkill: A Self-Evolving LLM Agent for Skill and Knowledge Accumulation in Traffic Simulation. arXiv preprint arXiv:2609.03753, 2026. URL https://arxiv.org/abs/2609.03753.

[140] Ziqiao Shang, Ling-Yue Ge, and Lan-Zhe Guo. GraphSkillAA: Attribution-Guided Skill-Graph Updating with Targeted Validation and Rollback. arXiv preprint arXiv:2609.20455, 2026. URL https://arxiv.org/abs/2609.20455.

[141] Ao Yan, Xin Zhang, Jiawei Du, and Joey Tianyi Zhou. SkillGLoW: Procedural-Family Skill Consolidation for Self-Improving Agents on Long-Horizon Task Streams. arXiv:2609.02217, 2026. URL https://arxiv.org/abs/2609.02217.

[142] Dongxiao Liu, Boren Zhu, Yuwen Ding, Linghui Li, Lei Li, and Xiaoyong Li. SkillRefine: Cross-Source Skill Induction and Execution Validation for LLM Agents in Refinery Planning Software. arXiv preprint arXiv:2609.30674, 2026. URL https://arxiv.org/abs/2609.30674.

[143] Yuxuan Liu, Zhaochen Su, Lingyun Xie, Yuhao Zhang, Qing Zong, Jiahe Guo, Zhongwei Xie, Yiyan Ji, Yauwai Yim, Hongyu Luo, Xiyu Ren, Ruan Chenyu, Haoran Li, and Yangqiu Song. SkillRevise: Improving LLM-Authored Agent Skills via Trace-Conditioned Skill Revision. arXiv:2606.01139, 2026. URL https://arxiv.org/abs/2606.01139.

[144] Haozhe Liu, Tian Ye, Sensen Gao, Qihang Cao, Yitong Li, Mingchen Zhuge, Duomin Wang, Ruihua Zhang, Ping Luo, Jiawang Bian, Lei Zhu, Ligeng Zhu, Enze Xie, and Song Han. SoL-Pi: Recursively Scaling Auto-Research Loops for Efficient Agent Harness. arXiv preprint arXiv:2609.20519, 2026. URL https://arxiv.org/abs/2609.20519.

[145] Taeckyung Lee, Rinat Amankos, Jeonghye Kim, Hyungjun Yoon, Woogyeol Jin, and Sung-Ju Lee. TISD: On-Policy Self-Distillation with Trajectory Intervention. arXiv preprint arXiv:2609.30878, 2026. URL https://arxiv.org/abs/2609.30878.

[146] Yicheng He, Chengsong Huang, Zongxia Li, Jiaxin Huang, and Yonghui Yang. VisPlay: Self-Evolving Vision-Language Models from Images. arXiv preprint arXiv:2511.15661, 2025. URL https://arxiv.org/abs/2511.15661.

[147] Mingxu Tao, Jiawei Hu, Xian Zhou, Wenpeng Hu, Jiajun Cheng, Yunbo Cao, Zhunchen Luo, and Guotong Geng. When Rules Learn: A Self-Evolving Agent for Legal Case Retrieval. arXiv:2606.17220, 2026. URL https://arxiv.org/abs/2606.17220.

[148] Liyan Tang, Cyrus Rashtchian, Chun-Sung Ferng, Andrew Tomkins, Da-Cheng Juan, and Tu Vu. WikiSkill: Compiling Agent Experience into Persistent Knowledge for Skill Evolution. arXiv:2608.27454, 2026. URL https://arxiv.org/abs/2608.27454.

[149] Junhao Zheng, Xidi Cai, Qiuke Li, Duzhen Zhang, ZhongZhi Li, Yingying Zhang, Le Song, and Qianli Ma. LifelongAgentBench: Evaluating LLM Agents as Lifelong Learners. arXiv:2505.11942, 2025. URL https://arxiv.org/abs/2505.11942.

[150] Yuxuan Cai, Yipeng Hao, Jie Zhou, Hang Yan, Zhikai Lei, Rui Zhen, Zhenhua Han, Yutao Yang, Junsong Li, Qianjun Pan, Tianyu Huai, Qin Chen, Xin Li, Kai Chen, Bo Zhang, Xipeng Qiu, and Liang He. Building Self-Evolving Agents via Experience-Driven Lifelong Learning: A Framework and Benchmark. arXiv:2508.19005, 2025. URL https://arxiv.org/abs/2508.19005.

[151] Parth Asawa, Christopher M. Glaze, Gabriel Orlanski, Ramya Ramakrishnan, Benji Xu, Asim Biswal, Vincent Sunn Chen, Frederic Sala, Matei Zaharia, and Joseph E. Gonzalez. Continual Learning Bench: Evaluating Frontier AI Systems in Real-World Stateful Environments. arXiv:2606.05661, 2026. URL https://arxiv.org/abs/2606.05661.

[152] Shuhan Xue, Zixin Ding, Yichen Shen, Yinjie Wang, Zhenfei Yin, Yingcheng Wu, Yuxin Chen, Mengdi Wang, and Ling Yang. PAST-Bench: Benchmarking the Foundations of Recursive Self-Improvement in Personal Agents. arXiv:2608.04003, 2026. URL https://arxiv.org/abs/2608.04003.

[153] Bo Deng, Kang Zhou, Lifan Guo, Chongyang Tao, Xuanren Chen, Chenggang Xie, Renzhao Liang, Feng Chen, and Chi Zhang. FinEvo-Bench: A Longitudinal Benchmark for Self-Evolving Agents in Professional Financial Workflows. arXiv:2608.06144, 2026. URL https://arxiv.org/abs/2608.06144.

[154] Ben Rank, Hardik Bhatnagar, Ameya Prabhu, Shira Eisenberg, Karina Nguyen, Matthias Bethge, and Maksym Andriushchenko. PostTrainBench: Can LLM Agents Automate LLM Post-Training? arXiv:2603.08640, 2026. URL https://arxiv.org/abs/2603.08640.

[155] Wanyi Chen, Xiao Yang, Xu Yang, Tianming Sha, Qizheng Li, Zhuo Wang, Bowen Xian, Fang Kong, Weiqing Liu, and Jiang Bian. Agent^2 RL-Bench: Can LLM Agents Engineer Agentic RL Post-Training? arXiv:2604.10547, 2026. URL https://arxiv.org/abs/2604.10547.

[156] Bohan Lyu, Yucheng Yang, Siqiao Huang, Jiaru Zhang, Qixin Xu, Xinghan Li, Xinyang Han, Yicheng Zhang, Huaqing Zhang, Runhan Huang, Kaicheng Yang, Zitao Chen, Wentao Guo, Junlin Yang, Xinyue Ai, Wenhao Chai, Yadi Cao, Ziran Yang, Kun Wang, Dapeng Jiang, Huan-ang Gao, Shange Tang, Chengshuai Shi, Simon S. Du, Max Simchowitz, Jiantao Jiao, Dawn Song, and Chi Jin. MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI. arXiv preprint arXiv:2605.08678, 2026. URL https://arxiv.org/abs/2605.08678.

[157] Feiyang Kang, Hanze Li, Adam Nguyen, Mahavir Dabas, Jiaqi W. Ma, Frederic Sala, Dawn Song, and Ruoxi Jia. Can Generalist Agents Automate Data Curation? arXiv:2606.04261, 2026. URL https://arxiv.org/abs/2606.04261.

[158] Zhangchen Xu, Junda Chen, Yue Huang, Dongfu Jiang, Jiefeng Chen, Hang Hua, Zijian Wu, Zheyuan Liu, Zexue He, Lichi Li, Shizhe Diao, Jiaxin Pei, Jinsung Yoon, Hao Zhang, Mengdi Wang, Radha Poovendran, Misha Sra, Alex Pentland, and Zichen Chen. AutoLab: Can Frontier Models Solve Long-Horizon Auto Research and Engineering Tasks? arXiv:2606.05080, 2026. URL https://arxiv.org/abs/2606.05080.

[159] Deyao Zhu, Xin Zhou, Shengling Qin, Xuekai Zhu, Hangliang Ding, Shu Zhong, Zixin Wen, Zhonglin Xie, Chenhui Gou, Linxuan Ren, Yueyang Wang, Junfeng Zhong, Rui Liu, Tian Gao, Yangguang Lin, Jingyuan Zhang, Maojia Song, Xuan Qi, Jinhong Wu, Chenyang Zhang, Yinzhu Piao, Ziru Niu, Hongbin Lin, Lingxiang Meng, Peng Tang, Chengyao Tang, Shanyu Wu, Huanyu Zheng, Yu Liu, Liya Zhu, He Wang, Ming Ding, Ziyu Wan, Hao Liu, Sibo Wang, Haotian Zhu, Xintian Zhang, Nan Chai, Yipeng Liu, Panhao Lai, Sihang Yuan, Zixin Su, Ge Zhang, Wangchunshu Zhou, Yantao Du, Wenhao Huang, and Guang Shi. EdgeBench: Unveiling Scaling Laws of Learning from Real-World Environments. arXiv:2607.05155, 2026. URL https://arxiv.org/abs/2607.05155.

[160] Yuanzhe Hu, Yu Wang, and Julian McAuley. Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions. arXiv:2507.05257, 2025. URL https://arxiv.org/abs/2507.05257.

[161] Varun Ursekar, Apaar Shanker, Veronica Chatrath, Yuan Xue, and Samuel Marc Denton. VeRO: A Harness for Agents to Optimize Agents. arXiv:2602.22480, 2026. URL https://arxiv.org/abs/2602.22480.

[162] Shanshan Zhong, Yi Lu, Jingjie Ning, Yibing Wan, Lihan Feng, Yuyi Ao, Leonardo F. R. Ribeiro, Markus Dreyer, Sean Ammirati, and Chenyan Xiong. SkillLearnBench: Benchmarking Continual Learning Methods for Agent Skill Generation on Real-World Tasks. arXiv:2604.20087, 2026. URL https://arxiv.org/abs/2604.20087.

[163] Xinyu Lu, Tianshu Wang, Pengbo Wang, zujie wen, Zhiqiang Zhang, Jun Zhou, Boxi Cao, Yaojie Lu, Hongyu Lin, Xianpei Han, and Le Sun. The Meta-Agent Challenge: Are Current Agents Capable of Autonomous Agent Development? arXiv:2606.04455, 2026. URL https://arxiv.org/abs/2606.04455.

[164] Congjie Zheng, Chuanyi Xue, Bin Liang, Jun Yang, and Changshui Zhang. SEAGym: An Evaluation Environment for Self-Evolving LLM Agents. arXiv:2606.17546, 2026. URL https://arxiv.org/abs/2606.17546.

[165] Yuyao Wang, Zhongjian Zhang, Mo Chi, Kaichi Yu, Yuhan Li, Miao Peng, Bing Tong, Chen Zhang, Yan Zhou, and Jia Li. EvoMemBench: Benchmarking Agent Memory from a Self-Evolving Perspective. arXiv:2605.18421, 2026. URL https://arxiv.org/abs/2605.18421.

[166] Yuhao Wu, Jingyuan Zhang, Jiajun Shi, Yuxuan Zhang, Xinping Lei, Junting Zhou, Zexuan Wang, Yuchen Wu, Huan Zhou, Duo Wang, Yinzhu Piao, Yongchang Peng, Yunfeng Shi, Jin Chen, Zuo Wang, Jinkai Liu, Jiaheng Liu, Wenxuan Zhang, Shen Yan, Wenhao Huang, and Ge Zhang. Aspire: Can Models Self-Evolve from Vague Goals? arXiv preprint arXiv:2608.31111, 2026. URL https://arxiv.org/abs/2608.31111.

[167] Proteus Authors. Proteus: A Harness-Agnostic Self-Evolution Framework for AI Agents, 2026. URL https://github.com/proteus-evolve/Proteus. Version v0.3.0; registry snapshot September 16, 2026.

[168] Human-Agent-Society. Reef: Continual Learning Infrastructure for Self-Improving Agents, 2026. URL https://github.com/Human-Agent-Society/reef. Version v0.0.2; registry snapshot September 16, 2026.

[169] AgentScope Team. ReMe: Memory Management Kit for Agents. Software, 2026. URL https://github.com/agentscope-ai/ReMe.

A Literature Catalogue

This appendix collects the recursive self-improvement methods, benchmarks, and software systems discussed in this survey. The method catalogue applies the unified taxonomy from Section 3; the benchmark catalogue summarizes the evaluation settings reviewed in Section 5.

A.1 Method papers

Table 2 records each method's choices across the nine dimensions and briefly describes its update mechanism. Multiple labels capture the resources and variants studied in each paper.

Table 2 Method papers and their classifications under the unified taxonomy.

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
A-MEM [21]	M	Teacher Sequential	On Event General	LLM feedback Other	No validation	Turns each new memory into a structured note and lets it revise summaries, attributes, and links of related historical memories in a Zettelkasten-style graph.
ACE [15]	C, M, S	Teacher Sequential	Off, On Trajectory, Batch General, Specialized	Train/dev set, Benchmark, Environment, Executable verifier, LLM feedback Binary, Non-binary, Ground truth, LLM-as-a-judge, Other	No validation	Turns execution feedback into incremental updates to a structured playbook, preserving useful strategies while revising and deduplicating experience for later tasks.
ACE (adversarial) [61]	P	Joint Sequential	Off Batch Specialized	Train/dev set, Executable verifier Binary, Non-binary	No validation	Updates solver and adversary adapters across rounds, using selected programs to train the solver and execution-derived preferences to train adversarial test generation.
ACON [19]	C	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Benchmark score, Combined metrics	Contrasts successful full-context runs with failed compressed-context runs to revise compression guidelines, selecting candidates by task success and context cost.
AHE [82]	H, C, M, S	Teacher Sequential	Off Batch General	Benchmark, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Artifact validation, Benchmark score	Distills batches of coding-agent trajectories into layered diagnostics, edits file-level harness components, and uses next-round task outcomes to verify or roll back each predicted change.
AIDE^2 [83]	H, C, M	Teacher Graph	Off Batch General	Train/dev set, Environment, Executable verifier Non-binary, Other	Benchmark score	Rewrites an AI research agent's code using prior candidate evaluations, retaining versions that improve research performance under fixed task budgets.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
AlgoEvo [29]	C, M, S	Self Graph	On Event, Batch General, Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Non-binary, Other	Instance result	Turns algorithm-search feedback into persistent experience cards and consolidates patterns supported by multiple tasks into reusable design skills for subsequent discovery.
AutoTTS [84]	H, C, M	Teacher Graph	Off Batch General	Train/dev set, Executable verifier Non-binary, Other	Combined metrics, Benchmark score	An LLM agent discovers test-time scaling controllers for other LLMs inside an environment built from pre-collected reasoning trajectories and probe signals, using execution-trace feedback to revise controller programs that then transfer to held-out benchmarks and model scales.
AWM [33]	M, S	Teacher Sequential	On, Off, Off→On Trajectory, Batch Specialized	Train/dev set, Environment Ground truth, Other	No validation	Induces reusable workflows from demonstrations or the agent's own completed web trajectories and retrieves them to guide later tasks.
CHIME [85]	C, M	Self Sequential	On, Off Trajectory Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, Other	Instance result	Separates planning and execution memories, attributes task outcomes before storing experience, and updates the relevant bank to guide later tasks without changing model weights.
COBRA-Skills [86]	C, S	Teacher Graph	Off Batch Specialized	Train/dev set, Environment, LLM feedback Binary, Non-binary, Other	Benchmark score	Uses a contextual bandit to allocate skill evaluations, then evolves a population through pruning, mutation, and crossover using observed task rewards.
Continual Harness [87]	P, H, C, M, S	Self, Teacher Sequential	On Step, Batch Specialized	Environment, LLM feedback Non-binary, LLM-as-a-judge, Other	No validation	Refines prompts, subagents, skills, and memory every few environment steps without resetting the episode, and optionally co-learns model weights from process-reward-labeled rollouts.
Cradle [88]	C, M, S	Joint Sequential	On Step, Event, Trajectory Specialized	Environment, LLM feedback Other	Instance result	A long-horizon computer-control agent that records observations and reflections, creates executable skills when reusable procedures emerge, and recalls them during later interaction.
DarwinX [89]	H, C, S	Teacher Graph	Off Batch General, Specialized	Benchmark, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Combined metrics	Branches and recombines harness versions, retaining complementary capabilities while using repeated evaluation and regression checks to control noisy improvements.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
DCE + SRCL [90]	P	Joint Sequential	Off Batch Specialized	Train/dev set, Executable verifier, LLM feedback Binary, Ground truth, Other	No validation	Co-evolves a student and its privileged self-teacher, combining on-policy distillation with verified concise rewrites to improve subsequent reasoning rounds.
DemoEvolve [91]	H, C	Teacher Graph	Off Batch Specialized	Train/dev set, Environment, Human Non-binary, Other	Benchmark score	Uses competent human trajectories as references so a coding proposer can diagnose sparse, noisy failures and produce more stable harness edits in long-horizon environments.
Designer-RSI [37]	C, M, S	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Benchmark score	Expands and repairs a graphic-design agent's procedural skill bank from user-task experience, admitting changes only when matched replays improve outcomes without regressing observed successes.
DGM [5]	H, C	Joint Tree	Off Batch General	Benchmark, Executable verifier Binary, Non-binary, Other	Combined metrics	Maintains an archive of coding-agent versions, samples parents, rewrites their harness code, and keeps empirically useful descendants as stepping stones for later evolution.
DIVE [92]	C, S	Self Graph	Off Batch Specialized	Train/dev set, Executable verifier Binary, Non-binary, Other	Benchmark score	Evolves diverse skill populations with a frozen model, combines skills across parents, and adapts the operators that generate new skills using measured utility.
DMoA [59]	O	Self Sequential	Off, On Batch Specialized	Train/dev set, LLM feedback Non-binary	No validation	Trains auxiliary routing weights from agent-output entropy so later queries use learned agent selection; the underlying language models remain frozen.
Dream-RSI [93]	H, C, M	Teacher Sequential	On Batch Specialized	Benchmark, Environment, Executable verifier Binary, Non-binary, Other	Combined metrics	Replays recorded discovery trees to evaluate and revise exploration-policy code cheaply, then redeploys the selected policy to guide fresh searches and expand the history available for improvement.
Dynamic Cheatsheet [94]	C, M, S	Teacher Sequential	On Trajectory General, Specialized	LLM feedback Other	No validation	Maintains a self-curated cheatsheet of strategies and code snippets across queries, updating memory from past attempts so later problems can reuse the accumulated experience.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
Ecdysis [40]	H, C	Teacher Sequential	Off Batch General	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Benchmark score	Turns batches of agent failures into repair specifications, edits executable runtime harnesses, and retains patches only when training-set performance improves.
EnvHarness [95]	C, M, S, O	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, Other	Artifact validation, Combined metrics, No validation	Revises environment components from rollouts, checks solvability and behavioral measures before retention, and stores extracted skills for subsequent interaction and environment revision.
Escher-Loop [96]	H	Teacher Graph	Off Batch General	Benchmark Non-binary	Benchmark score	Maintains populations of task agents and optimizer agents; optimizers revise task programs and themselves while task-agent scores provide the signal for optimizer evolution.
Evo-Harness [26]	C, S	Joint Sequential	On Trajectory Specialized	Train/dev set, LLM feedback Binary, Other	Instance result	After each one-shot task, a solver exposes candidate experience and a separate evolver compiles useful context into structured harness skills for later tasks.
EvolveNet [97]	H, C	Teacher Graph	Off Batch General, Specialized	Train/dev set, Executable verifier, LLM feedback Binary, Non-binary, Other	Benchmark score	Evolves a shared harness on separate local workloads, combines evidence-backed changes with explicit scopes, and checks regressions before broadcasting the next version.
EvolveTrade [98]	C, S	Teacher Sequential	On Batch Specialized	Environment, LLM feedback Non-binary, Other	No validation	Periodically rewrites a trading agent's textual policy using its decisions, tool calls, rationales, and realized returns, carrying the revised instructions into later trading periods.
EvoOntology [99]	H, C, M	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, Other	Benchmark score	Evolves a data agent's stored content, ontology schema, and executable tool layer from grounded probes and task traces, retaining changes through paired validation.
EvoSkill-GUI [100]	C, M, S	Self Sequential	On Event, Trajectory Specialized	Environment, LLM feedback Binary, LLM-as-a-judge, Other	Instance result	Revises reusable GUI skill packages during execution and after trajectory-level critique, storing improved procedures and recovery experience for similar later tasks.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
EvoTrainer [39]	P, H, M, S, O	Teacher Graph	Off Batch General, Specialized	Train/dev set, Benchmark, Environment, Executable verifier, LLM feedback, Human Binary, Non-binary, Other	Combined metrics	Uses rollouts to revise training recipes and diagnostic code, stores reusable lessons and skills, and promotes policy-training versions through performance and behavioral checks with human confirmation.
Experience Funnel [101]	P, C, M, S	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, LLM feedback Binary, Non-binary, Other	Benchmark score	Alternates validation-gated updates to a textual experience state with policy distillation, carrying both the improved state and student weights into the next round.
ExpGraph [102]	P, C, M, S	Teacher Sequential	Off Trajectory General, Specialized	Train/dev set, Environment, Executable verifier Binary, Non-binary, Other	Instance result	Converts task histories into a maintained experience graph and learns retrieval from measured gains over no-memory runs, while leaving the task executor frozen.
ForeDreamer [103]	H, C, M, S	Teacher Graph	Off Batch Specialized	Train/dev set, Environment, Executable verifier Binary, Non-binary, Other	Artifact validation, Benchmark score	Evolves forecasting lessons and evidence-processing instructions and tools, validating updates and reusing existing tools across procedural branches.
G-Zero [104]	P	Joint Sequential	Off Batch General	LLM feedback Non-binary	No validation	Co-evolves a Proposer and a Generator for open-ended generation: the Proposer is trained with GRPO to find queries and hints on which the Generator's hint-conditioned response shifts most, and the Generator is trained with DPO to internalize the hints, with no external judge.
GEA [105]	H, C	Joint Graph	Off Batch General	Benchmark, Executable verifier, LLM feedback Binary, Non-binary, Other	Artifact validation, Combined metrics	Pools code changes, execution traces, and outcomes across a group of archived agents, then uses the shared evidence to evolve each member's harness.
GEPA [18]	C	Teacher Graph	Off Batch Specialized	Train/dev set, Executable verifier, LLM feedback Binary, Non-binary, Ground truth, Other	Benchmark score	Reflects on execution traces and task feedback to revise prompts, then evaluates and recombines complementary candidates through Pareto-based search.
Growing Harness [106]	H, C	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, Executable verifier Binary, Non-binary, Other	Artifact validation, Benchmark score	Grows a shared executable harness from failure-localized code edits, jointly repairs task windows, and rolls back changes that harm held-out success.
Gödel Agent [107]	H, C	Self Sequential	Off Batch General	Benchmark, LLM feedback Non-binary, Other	Benchmark score	Lets an agent inspect and rewrite its own executable logic, including the logic used for later self-modification, and empirically selects better descendants.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
Harness-of-Harness [108]	C, M, O	Joint Sequential	On Trajectory Specialized	Environment, Executable verifier, LLM feedback Binary, LLM-as-a-judge, Other	Artifact validation	Runs repeated planning, development, and QA cycles over multiple days, preserving project state and verification evidence so later iterations can improve the same software project.
Harness-Zero [109]	H, C, M, S	Teacher Sequential	Off Batch Specialized	Train/dev set, Benchmark, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Artifact validation, Instance result	Revises harness tools, middleware, skills, and memory from recurring failures, then adapts the evolved harness into reference resources that guide student execution.
HarnessBank [110]	H, C	Teacher Graph	Off Batch General, Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Artifact validation, Combined metrics	Maintains a semantically diverse bank of high-performing harnesses, reinvents or recombines their mechanisms, and admits offspring only after gated verification.
HarnessEvolve [111]	H, C, S	Teacher Tree	Off Batch Specialized	Train/dev set, Environment, LLM feedback Binary, Non-binary, Ground truth, LLM-as-a-judge, Other	Artifact validation, Combined metrics	Compares failed runs with verified reference trajectories to localize root causes, then accepts harness edits only after quality and recent-performance gates.
HarnessFix [38]	H, C, M	Teacher Sequential	Off Batch Specialized	Benchmark, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Artifact validation, Benchmark score	Compiles failed trajectories into a harness-aware representation, attributes failures to responsible steps and components, and applies scoped patches under regression-aware validation.
HarnessPAI [112]	H, C, M, S	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Artifact validation, Benchmark score	Revises robot execution programs from rollout videos and diagnostics, validates repairs across development scenes, and stores successful repairs as reusable skills.
HarnessX [113]	P, H, C, M	Teacher Sequential	Off Batch General, Specialized	Benchmark, Environment, Executable verifier Binary, Ground truth, Other	Artifact validation, Benchmark score	Composes typed harness primitives, evolves prompts, tools, memory, and control flow from execution traces, and can turn the resulting trajectories into model-training signals.
HeLa-Mem [114]	M	Teacher Sequential	On Event, Trajectory General	Environment, LLM feedback Other	No validation	Uses Hebbian co-activation to evolve an episodic memory graph and periodically distills dense memory hubs into reusable semantic knowledge.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
HGM [31]	H, C	Self Tree	Off Batch General	Benchmark, Executable verifier, LLM feedback Binary, Non-binary, Other	Benchmark score	Uses descendant performance to estimate a coding agent's self-improvement potential, guiding which archived agents rewrite themselves and which receive further evaluation.
Hyperagents [17]	H, C, M	Teacher Graph	Off Batch General, Specialized	Benchmark, Executable verifier, LLM feedback Binary, Non-binary, Other	Artifact validation, Benchmark score	Makes both the task agent and its meta-agent editable, allowing archived agents to improve task-solving code and the procedure used to generate later modifications.
HyperSkill [115]	C, M, S	Teacher Sequential	On Batch Specialized	Environment, LLM feedback Non-binary, Other	Instance result	Represents subtasks and reusable skills in a trajectory hypergraph, retrieves along two structural paths, and periodically prunes or merges skills using observed utility.
ISM [116]	M, S	Teacher Sequential	On Trajectory, Batch Specialized	Benchmark, Executable verifier, LLM feedback Binary, Other	Artifact validation, Instance result	Maintains a verified bank of mathematical strategies, using successes and failures to revise, merge, reinforce, and prune memories across isolated task episodes.
JAZ [117]	H, C, M, S	Teacher Sequential	On Trajectory Specialized	Benchmark, Environment, Executable verifier Binary, Non-binary, Other	No validation	Lets a meta-agent revise a solver's prompts and executable skills from task feedback, while a programmable context interface also supports long-horizon memory.
KSI [24]	C, M	Joint Graph	On, Off Trajectory, Batch General, Specialized	Benchmark, Train/dev set, Environment, Executable verifier, LLM feedback Binary, Ground truth, Other	Artifact validation	Keeps task agents disposable while evolving a shared knowledge base through evidence-grounded task forums, cross-task discussion, and distillation into guidance for later agents.
MCE [118]	H, C, M, S	Teacher Graph, Sequential	Off, On Batch, Trajectory Specialized	Train/dev set, Benchmark, LLM feedback Binary, Non-binary, Ground truth, Other	Artifact validation, Benchmark score	A meta-agent combines historical context-engineering skills, while a base agent applies them to revise reusable context files and retrieval code from task feedback; validation selects the best context.
MediSkill-Evo [119]	C, M, S	Teacher Sequential	Off Event, Trajectory Specialized	Train/dev set, Environment, Executable verifier Other	Artifact validation, Instance result	Writes clinical interaction experience into four typed knowledge banks only after provenance, scope, and process checks, then constrains later actions with the validated knowledge.
Mem^{2}Evolve [76]	H, C, M, S	Teacher Sequential	On, Off→On Trajectory, Batch General	Benchmark, LLM feedback Non-binary, Ground truth, LLM-as-a-judge	Instance result	Co-evolves an experience memory and an asset memory: prior lessons guide the creation of tools or expert agents, whose use produces new experience for the next round.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
Memp [120]	C, M, S	Teacher Sequential	Off→On Trajectory Specialized	Train/dev set, Environment Other	Instance result	Distills trajectories into step-level instructions and script-like procedures, then continuously adds, corrects, and deprecates entries as experience accumulates.
MemRL [3]	C, M, S	Teacher Sequential	On, Off Trajectory Specialized	Train/dev set, Benchmark, Environment, Executable verifier, LLM feedback Binary, Non-binary, Other	No validation	Stores trajectory-derived experiences with utility values, updates those values from task rewards, and combines utility with semantic relevance when retrieving guidance for later tasks.
Mendel Gödel Machine [32]	H, C	Joint Graph	Off Batch General	Benchmark, Executable verifier, LLM feedback Binary, Non-binary, Other	Combined metrics	Improves coding-agent descendants using evidence from several tasks or from another lineage, adding reaction-norm mutation and cross-lineage hybridization to archive search.
Meta-Harness [121]	H, C, M	Teacher Graph	Off Batch Specialized	Benchmark, Executable verifier Binary, Non-binary, Other	Combined metrics	Lets an agentic proposer inspect the code, scores, and execution traces of all prior candidates through a filesystem, then search for better task-specific harness code.
MM-Zero [122]	P	Joint Sequential	Off Batch General	Executable verifier, LLM feedback Binary, Non-binary	No validation	Trains Proposer, Coder, and Solver roles from one base VLM with GRPO: the Proposer poses visual concepts and questions, the Coder renders them into images as executable code, and the Solver reasons over them, so multimodal training data is generated from zero seed images.
ModularRSI [66]	H, C, M	Self Graph	Off Batch General	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Other	Artifact validation	Separately evolves five harness modules from execution experience, checks candidate patches for validity, and integrates the retained modules into a general-purpose agent.
MoMHa [123]	H, C	Teacher Graph	Off Batch Specialized	Train/dev set, Executable verifier, LLM feedback Binary, Non-binary, Other	Artifact validation, Combined metrics	Uses historical harness code and execution traces to propose new harnesses, retaining candidates that improve a joint accuracy, safety, and token-cost objective.
MOSS [124]	H	Teacher Sequential	On Batch General, Specialized	Environment, Executable verifier, LLM feedback, Human Non-binary, LLM-as-a-judge, Other	Artifact validation, Combined metrics	Turns deployment failures into source-code revisions, tests rebuilt agents on captured tasks, and promotes fixes through consent and health-check gates.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
NavHarness [125]	C, M, S	Joint Sequential	On Step, Event, Trajectory, Batch Specialized	Environment, LLM feedback Non-binary, LLM-as-a-judge, Other	No validation	Carries navigation experience across fresh sessions through revisable spatial memory, task handovers, and consolidated house notes and skills, using observations and model-based completion checks.
NeoHorse-1 [126]	P	Teacher Sequential	Off Batch General	Train/dev set, Environment, LLM feedback Non-binary, Ground truth, Other	No validation	Uses routing and verification signals to organize agent experience for post-training, then reallocates the next training mixture according to the updated model's capability gaps.
OpenSkill [127]	H, C, S	Teacher Sequential	Off Batch Specialized	Environment, Executable verifier, LLM feedback Binary, Non-binary, Other	No validation	Builds skills and virtual tests from open-world resources, then uses execution diagnostics and targeted retrieval to refine skills before target-task use.
P^{2}O [128]	P, C	Self Sequential	Off Batch General, Specialized	Benchmark, Executable verifier, LLM feedback Binary, Ground truth, Other	Benchmark score	A self-improvement loop that alternates policy and prompt optimization: evolved prompts give policy learning better exploration templates, while the improving policy supplies increasingly difficult, high-value seeds for the next prompt round; prompt-induced gains are distilled into the model parameters.
PANDO [129]	H, C, M, S	Teacher Sequential	On Trajectory Specialized	Environment, LLM feedback Binary, Other	Instance result	Distills completed web interactions into reusable rules and executable routines, then updates confidence, merges redundant skills, and demotes unreliable ones during continued use.
PILOT [34]	C, M, S	Teacher Graph	On Event, Batch General, Specialized	Environment, LLM feedback, Benchmark Binary, Other	No validation, Instance result	A live supervisor steers active workers and updates reusable skills and memory during execution; successful runs contribute their updates to the shared state used in later iterations.
Prime Agent [130]	C, M, S	Self Sequential	On Step, Event, Trajectory General, Specialized	Environment, Executable verifier Other	Instance result	A persistent RLM-style harness that preserves histories, memories, skills, prompts, and subagent specifications across trajectories so later work can build on earlier context.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
Procedural Graphs [131]	C, M, S	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, LLM feedback Binary, Non-binary, Other	Artifact validation, Benchmark score	Builds a procedural graph from successful and failed trajectories, validates graph edits before retaining them, and turns relevant subgraphs into step-level guidance for later tasks.
R-Few [132]	P	Joint Sequential	Off Batch General	Train/dev set, LLM feedback, Human Binary, Non-binary	No validation	Guides Challenger–Solver self-play with lightweight human oversight: the Challenger samples a few human-labeled examples to ground question generation, and the Solver trains on mixed human and synthetic data under an adaptive difficulty-based curriculum, which mitigates drift and diversity collapse.
R-Zero [133]	P	Joint Sequential	Off Batch General	LLM feedback Binary	No validation	Initializes a Challenger and a Solver from one base model; the Challenger is rewarded for questions at the edge of the Solver's ability and the Solver trains on majority-vote pseudo-labels, so the curriculum tracks the Solver across rounds without any external data.
ReasoningBank [20]	C, M	Joint Sequential	On Trajectory General, Specialized	Environment, LLM feedback Binary, Other	No validation	Distills generalizable reasoning strategies from self-judged successes and failures, retrieves them for later tasks, and uses additional test-time rollouts to improve the memory bank.
Recuris [30]	C, M, S	Teacher Sequential	Off, Off→On Trajectory Specialized	Environment, Executable verifier, LLM feedback Other	Artifact validation, Instance result	Couples working and experiential memory so long-horizon execution localizes memory failures, then lets a fixed meta-agent make validation-gated updates to skill memory.
Red Queen Gödel Machine [134]	H, C	Teacher Tree	Off Batch General	Benchmark, LLM feedback Non-binary, LLM-as-a-judge	Combined metrics	Extends archive-based self-improvement so the agent and its utility or evaluator can both evolve, while preserving stable selection within each evaluation epoch.
RHI [135]	C	Teacher Sequential	Off Batch Specialized	Benchmark, LLM feedback Non-binary, LLM-as-a-judge	Benchmark score	Represents an agent loop as a prompt-level harness and repeatedly revises it using pairwise feedback accumulated across consecutive versions.
RISE [47]	P	Self Graph	Off Batch General, Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, Other	No validation	Alternates verifiable-reward training with distillation from a teacher extrapolated from its own checkpoints, refreshing the teacher as the student improves.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
RRSI [10]	H, C, M, S	Teacher Graph	Off Batch General	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Combined metrics	Regularizes harness evolution with shrinking edit budgets, history-guided proposals, leakage checks, and cost-aware selection to reduce overfitting to evolution tasks.
RSIAgent [136]	C, M, S	Self Sequential	Off Trajectory, Batch Specialized	Environment, LLM feedback Binary, LLM-as-a-judge, Other	Instance result	Explores unfamiliar environments with curriculum, actor, and verifier roles, updates reusable memory from verified experience, and freezes that memory for downstream execution.
SAGEAgent [23]	C, M, S	Teacher Sequential	Off Trajectory, Batch Specialized	Train/dev set, Environment, LLM feedback Non-binary, Other	Artifact validation	Accumulates decision episodes and periodically revises explicit acquisition rules using cost and uncertainty feedback, improving which diagnostic modalities a frozen agent requests.
SEAL [2]	P	Self Sequential	Off Batch General	Train/dev set Non-binary, Ground truth	Benchmark score	The model writes self-edits that specify its own finetuning data and update directives; downstream performance rewards self-edits that produce useful weight changes.
SEEK [55]	C, S	Teacher Sequential	On Batch Specialized	Train/dev set, Human, LLM feedback Non-binary, Ground truth, Other	Benchmark score	Uses reviewed search-evaluation errors to revise a skill bank, accepting changes only when they improve new cases without degrading historical replay performance.
Self-Harness [137]	H, C, M	Self Graph	Off Batch Specialized	Benchmark, Executable verifier, LLM feedback Binary, Non-binary, Other	Combined metrics	Uses the same fixed model to mine recurring failures, propose bounded edits to its own harness, and merge only candidates that pass held-in and held-out regression tests.
SePO [138]	C	Self, Teacher Tree	Off Batch General, Specialized	Train/dev set, Executable verifier Binary, Non-binary	Benchmark score	Evolves a prompt agent's own system prompt on a multi-task pool, then uses that improved prompt optimizer to evolve task-agent prompts for target tasks.
SIFT [35]	H, C	Teacher Tree	Off Batch General	Benchmark, Executable verifier, LLM feedback Binary, Non-binary, Other	Combined metrics	Searches a tree of coding-agent versions using pairwise code judgments alongside task evaluations, reducing the number of expensive benchmark runs needed to choose promising branches.
SimSkill [139]	H, C, M, S	Teacher Sequential	Off, On Trajectory Specialized	Environment, Executable verifier, LLM feedback Binary, LLM-as-a-judge, Other	Artifact validation, Instance result	Explores traffic-simulation tasks through an action–critic loop and consolidates completed episodes into reusable code skills and linked knowledge pages.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
SkillAA [140]	C, M, S	Teacher Sequential	Off Batch Specialized	Train/dev set, Executable verifier, LLM feedback Binary, Non-binary, Ground truth, Other	Benchmark score	Attributes task failures to nodes in a skill graph, proposes local edits, and applies targeted and full-pool validation with rollback before retaining an update.
SkillAdam [4]	C, M, S	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Combined metrics	Revises skill documents using a persistent issue tracker and an adaptive edit budget, retaining changes when batch-level evaluation shows gains without unacceptable regressions.
SkillEvoReg [51]	H, C, S	Teacher Sequential	Off, On Trajectory, Batch Specialized	Train/dev set, Benchmark, Executable verifier, LLM feedback Binary, Non-binary, Other	Artifact validation, Instance result, Benchmark score	Regularizes repeated skill updates using skill dropout, localized complexity control, and paired counterexample checks to reduce overfitting and regressions.
SkillGLoW [141]	C, M, S	Teacher Graph	On, Off Trajectory, Batch Specialized	Benchmark, Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, Other	Benchmark score	Groups task-local experience by shared solving procedure, consolidates it into reusable skill priors, and checks library revisions through execution before combining them with fresh local skills.
SkillHEX [36]	C, S	Teacher Graph	On Trajectory Specialized	Benchmark, Environment, Executable verifier, LLM feedback Binary, Non-binary, Other	Artifact validation, Instance result	Tests explicit failure hypotheses against cached outputs and shares the resulting evidence across skill-revision branches to guide budget-limited test-time improvement.
SkillPivot [27]	C, S	Teacher Sequential	On Batch Specialized	Benchmark, Environment, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Combined metrics	Locates where a failed trajectory went off track, compares it with a teacher's continuation from the same prefix, and retains localized skill edits after preservation and generalizability checks.
SkillProx [57]	C, M, S	Teacher Graph	Off Batch Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Artifact validation, Combined metrics	Revises skills from execution feedback, then removes or consolidates low-utility content through edits checked for structural validity, lower complexity, and preserved task performance.
SkillRefine [142]	C, M, S	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, Ground truth, Other	No validation	Builds refinery-planning skills from documentation and expert records, then uses execution constraints and reference comparisons to repair skill content and retrieval metadata.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
SkillRevise [143]	C, S	Teacher Sequential	On Trajectory Specialized	Benchmark, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Combined metrics	Diagnoses skill defects from execution traces and verifier feedback, revises the skill with repair principles, and returns the best tested candidate within a bounded budget.
SkillSmith [28]	H, M, S	Teacher Graph	Off Batch General	Train/dev set, Executable verifier, LLM feedback Non-binary, Other	Artifact validation, Combined metrics	Reflects on failures to jointly edit skills and executable tools, validates bundled changes progressively, and records anti-patterns to prevent repeated mistakes.
SoL-Pi [144]	H, C	Teacher Graph	Off Batch General	Train/dev set, Environment, Executable verifier, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	Combined metrics	Scales parallel research loops over harness changes, validates efficiency gains under capability constraints, and combines retained mechanisms into a lower-cost agent harness.
TextGrad [9]	C	Teacher Sequential	Off Batch Specialized	Train/dev set, Executable verifier, LLM feedback Binary, Non-binary, Ground truth, LLM-as-a-judge	Benchmark score	Backpropagates textual feedback through a computation graph to revise reusable prompts, retaining updates that improve validation performance.
TISD [145]	P	Joint Sequential	Off Batch Specialized	Train/dev set, Executable verifier, LLM feedback Binary, Other	No validation	Uses teacher–student disagreement to branch a student rollout, regenerates its continuation, and distills the resulting trajectories while updating the teacher from the student.
TRACE [25]	C, S	Teacher Sequential	Off Batch Specialized	Benchmark, LLM feedback Binary, Other	Benchmark score	Groups trajectories by invoked skills and contrasts successful with failed behavior after each development evaluation round to rewrite a reusable behavioral skill bank.
TTCS [53]	P	Joint Sequential	On Batch Specialized	LLM feedback Binary	No validation	Co-evolves a question synthesizer and a solver during test-time training: the synthesizer generates progressively harder variants of the test questions guided by solver feedback, and the solver updates its weights with self-consistency rewards on both original and synthesized questions.
TTSE [22]	C, M, S	Teacher Sequential	Off, On Batch, Trajectory Specialized	Train/dev set, Benchmark, Environment, LLM feedback Binary, Non-binary, LLM-as-a-judge, Other	No validation, Instance result	Separates environmental facts from conditional procedures, then uses task feedback to retire misleading rules, resolve contradictions, and update both banks for later tasks.

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Table 2 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Method	**Targets**	**Organization** Updater / topology	**Schedule and scope** Mode / frequency / scope	**Feedback** Source / type	**Acceptance**	**Update mechanism**
VisPlay [146]	P	Joint Sequential	Off Batch General	LLM feedback Binary	No validation	Assigns one base VLM the roles of an image-conditioned Questioner and a multimodal Reasoner, jointly trained with GRPO using diversity and difficulty rewards, so the model improves visual reasoning from unlabeled images alone.
When Rules Learn [147]	C, S	Self Sequential	Off Batch Specialized	Train/dev set, Executable verifier Non-binary	Benchmark score	An agent creates query-rewriting rules, plans experiments over rule combinations, and removes rules that fail to improve legal retrieval metrics.
WikiSkill [148]	C, M, S	Teacher Sequential	Off Batch Specialized	Train/dev set, Environment, LLM feedback Binary, Non-binary, Ground truth, Other	Benchmark score	Maintains an immutable raw-trace layer and a persistent wiki that survives rejected skill edits, allowing later skill updates to build on accumulated evidence.

A.2 Benchmarks

Table 3 summarizes the evaluation settings of the reviewed benchmarks. It records the resources being updated, how improvement and testing are organized, the feedback available for improvement, and the reported measures. Task performance includes success rate, reward, and output quality; gain measures improvement over a reference system. Cost and time cover resource use and elapsed time, respectively.

Table 3 Benchmarks and their evaluation settings.

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Benchmark	Resources	Mode	Task setup	Feedback for updates	Reported measures
LifelongAgentBench [149]	C, M	On	Skill-dependent task sequences in interactive environments	Environment observations and terminal success/failure	Task performance, Gain, Cost
StuLife [150]	P, C, S, M	On	Stateful campus task sequence; includes a fine-tuning baseline	Environment observations, tool results, and task outcomes	Task performance, Gain, Interaction turns
Evo-Memory [68]	C, M	On	Task streams with varied order and difficulty	Task trajectories and correctness signals	Task performance, Gain, Interaction steps
SkillFlow [69]	C, S	On	Task families ordered by within-family difficulty	Execution traces and rubric-based verifier feedback	Task performance, Gain, Cost
CL-Bench [151]	C, M, O	On	Related instances in shared environments, including concept drift	Environment observations and task-dependent feedback	Task performance, Gain, Cost

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Table 3 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Benchmark	Resources	Mode	Task setup	Feedback for updates	Reported measures
AgentStream [70]	C, S, M	On	Isolated, sequential, and interleaved domain streams	Execution outcomes and self-evaluation	Task performance, Gain, Cost
PATH-Bench [71]	C, S, M	On	Repeated probes within helpful or interfering task histories	Intervening-task experience; probes do not update state	Task performance, Transfer, Forgetting
PAST-Bench [152]	S, M	On	Learn, reuse, and control episodes in fresh sessions	Rules, procedures, and corrections in learning episodes	Task performance, Gain
ContinualSkillBench [77]	C, S	On	Tasks ordered by difficulty and skill dependencies	Execution trajectories and evaluator feedback	Task performance, Gain
FinEvo-Bench [153]	S, M	On	Shuffled, interleaved streams of related professional cases	Rubric-based feedback after task completion	Task performance, Gain, Cost
PostTrainBench [154]	C, O	On	Research-agent loop over target-model post-training; final checkpoint test	Execution logs and agent-run experiments	Task performance, Gain, Cost, Time
Agent^2 RL-Bench [155]	C, O	On	Research-agent loop with repeated training and target-model submissions	Grading-server scores for submitted models	Task performance, Gain, Cost, Time
MLS-Bench [156]	C, O	On	Research-agent loop for ML-method components across evaluation settings	Training and visible-test metrics returned by experiment tools	Task performance, Gain, Cost
Curation-Bench [157]	C, O	On	Research-agent loop over data-curation policies for a target model	Scores from fixed training and evaluation pipelines	Task performance, Gain, Cost
AutoLab [158]	C, O	On	Iterative improvement of a working implementation within a time budget	Execution results and local benchmark measurements	Task performance, Gain, Cost, Time
EdgeBench [159]	C, O	On	Repeated attempts within a long-running task	Local tests and submission-gated judge feedback	Task performance, Gain, Cost, Time
RSIBench-Data [78]	C, O	On	Research-agent loop revising training-data strategies for a target model	Checkpoint scores, trajectories, and execution diagnostics	Task performance, Gain, Cost
MemoryAgentBench [160]	C, M	Off	Incremental context ingestion, then question answering	Provided context chunks and labeled examples	Task performance, Gain, Cost, Time
MemoryBench [72]	P, M	Off	Feedback training, then held-out tests; includes weight fine-tuning	Simulated user comments and actions	Task performance, Gain, Cost, Time
VeRO-Bench [161]	H, C	Off	Optimize target harnesses, then evaluate held-out tasks	Development scores and structured execution traces	Task performance, Gain, Cost
SkillLearnBench [162]	S	Off	Generate skills from seed tasks, then test held-out instances	Seed executions and self/teacher feedback, depending on method	Task performance, Gain, Cost
Meta-Agent Challenge [163]	H, C	Off	Agent development followed by held-out verification	Development-set correctness and aggregate scores	Task performance, Cost, Time
SEAGym [164]	H, C, M, S	Off	Training-task updates; frozen-snapshot validation and transfer tests	Training trajectories and visible verifier feedback	Task performance, Gain, Cost, Time
EvoAgentBench [79]	S	Off	Training and held-out tasks linked by shared procedural abilities	Training trajectories and verifier outcomes	Task performance, Gain, Cost
GDPevo [73]	S	Off	Training rules recombined in held-out workflows	Reference answers, grader scores, or environment observations	Task performance, Gain, Cost

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Table 3 (continued)

P: parameters; H: harness code; C: context; M: memory; S: skills; O: other artifacts. Off/On: offline/online.

Benchmark	Resources	Mode	Task setup	Feedback for updates	Reported measures
HarnessOpt-Bench [74]	H, C	Off	Select a revised harness, then test held-out examples	Development traces/results and aggregate validation scores	Task performance, Gain, Cost
Evo-Bench [81]	H, C	Off	Harness-sensitive tasks split for optimization and final testing	Optimization scores, trajectories, and diagnostics	Task performance, Gain, Cost
S3Gym [75]	P, C, M	Off	Game exploration followed by stricter held-out evaluation	Environment observations and self-judgments	Task performance, Gain
HarnessDev [80]	H, C	Off	Harness creation/evolution before testing frozen code	Development cases and downstream execution feedback	Task performance, Gain, Cost
EvoMemBench [165]	C, M	On, Off	Online within/across episodes; frozen-memory transfer across environments	Incoming information, tool results, and prior trajectories	Task performance, Gain, Cost
Aspire [166]	P, H, C, M	On, Off	Final-only and adaptive-feedback protocols; weight or harness tracks	Agent-created validation data; aggregate scores in adaptive protocol	Task performance, Gain, Cost, Time

A.3 Software systems

Table 4 Software systems for agent self-improvement.

System	Main functionality	Repository
Proteus [167]	Evolves agent harnesses, records version snapshots, and measures changes.	`https://github.com/proteus-evolve/Proteus`
Reef [168]	Connects execution records and feedback to updates, evaluation, and version deployment.	`https://github.com/Human-Agent-Society/reef`
ReMe [169]	Stores, retrieves, and maintains agent memories for reuse across tasks.	`https://github.com/agentscope-ai/ReMe`